System Analysis And Design

CSE-3411

CareerBridge Pro



Group Name: Bridgify

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Chapter 1

Introduction

The rapid changing world of work presents challenges for both students and employers. The uncertainty about which skills to master can be a source of stress for students, while employers often face a shortage of candidates with the precise qualifications.

This project presents an intelligent system that bridges this gap between learning and opportunities. This system helps students to determine their skills gaps, to learn the needed skill, to plan their future career and to track their progress, whereas it allows employers to search for qualified candidates more efficiently.

1.1 Purpose

The purpose of this project is to guide students towards the proper skills and to provide a career path from education to employment. Additionally, this project aims to assist employers by making the process of searching for the appropriate candidates faster and easier. In summary, the system should reduce the gaps between education and industry demand.

1.2 Problem Statement

The main purpose of this project is to guide students towards the proper skills and to provide a career path from education to employment. Additionally, this project aims to assist employers by making the process of searching for the appropriate candidates faster and easier. In summary, the system should reduce the gaps between education and industry demand.

1.2.1 Problem Background

- Lack of clear skill direction: The overwhelming number of learning resources available, students find it confusing which skill should they learn to enhance their career

progression.

- High job market competition: Graduates are increasing every year, which causes extreme competition.
- Difficulty in finding ideal candidates: Employers need to struggle in searching for candidates with specific skills required by the job description.

1.2.2 Problem Description

- Lack of skill awareness: A lot of students are not aware of skills that are crucial to their careers, and will therefore learn less useful subjects.
- No clear career roadmap: Students do not have a clear plan on how to learn and get a job.
- Gap between theory and practical skills: Academic learning has mostly a theoretical approach whereas industry expects a more practical experience and skills.
- Job matching problem: A student cannot easily find job descriptions and openings that perfectly matches his/her abilities and skills whereas an employer cannot effectively find employees with required skills.

1.2.3 Problem Reasoning

- Absence of a unifying platform: There is no single platform which will integrate learning, skill development and job matching all together.
- Traditional educational approach: Universities and colleges have historically based education more on theory rather than practical knowledge and skill sets.
- Lack of proper guidance: Students rarely get the proper guidance while choosing their career path and deciding on what skills they should acquire.
- Traditional recruitment methods: Traditional recruitment techniques by employers is a time-consuming and inefficient process.

1.3 Goal

The goal of this project is to develop a smart system that helps students to acquire necessary skills for their career and to provide a precise learning roadmap toward them as well as connects learning and job opportunities. It also aims to enable employers to easily search and identify ideal candidates, saving them time and efforts.

1.4 System Development Life Cycle (SDLC)



Figure 1.1: Software Development Life Cycle (SDLC) for CareerBridge Pro

Planning: Identify the needs and scope for a smart career system which matches learning and employment opportunities; Define the key features: Smart profile system, AI Skill gap analyzer, Career path planner, performance tracking system.

Requirements Gathering: We will collect requirements from students and job seekers, so we know exactly what the system requires, e.g. Recommend skills, clear learning pathways, matching the user to suitable jobs and the progress tracker functionality etc.

Design: The architecture of the system and the UI/UX design will be developed; including the layouts for the user profiles, skill analysis dashboard, the career roadmap view and the performance tracker.

Development: The system will be developed and some of the core functionality included in the system will be authentication, profile handling, AI driven skill gap analysis, career path generation, job matching and the performance tracker.

Testing: All of the features will be tested and ensure it is performing accurately. We will test for skill analysis accuracy, career path suitability, profile handling system accuracy and general performance accuracy.

Deployment: The CareerBridge Pro system is an online system. It will be accessible through the internet or an application on user's devise. Thus, both students and employees would easily be able to log on to and use the CareerBridge Pro system.

Maintenance: We are also going to be responsible of maintaining the system through constant debugging, performance improvement and upgrade with new features based on the feedback from users.

Chapter 2

System Study on Information Gathering

2.1 Introduction

This chapter documents the systematic approach employed to collect, analyze, and synthesize information from diverse sources throughout a comprehensive three-month research period.

The information gathering process forms the foundation of Bridgify requirements specification, ensuring that the system addresses real-world needs of all stakeholders in the job-market ecosystem.

The methodology involved multiple complementary approaches:

- Primary Research: Direct surveys, structured interviews, and field observations
- Secondary Research: Academic literature review, competitive analysis, and industry reports
- Technical Investigation: Technology evaluation, architecture research, and best practices analysis

This multi-faceted research approach enabled comprehensive understanding of:

- Current challenges faced by job seekers and learners
- Existing gaps in job and skill development platforms
- User expectations and behavior patterns
- Market trends and technological opportunities

The findings obtained through this systematic information gathering directly informed functional requirement definitions, feature prioritization decisions, and fundamental system design choices that shape the Bridgify platform architecture and user experience.

2.2 Information Sources

Comprehensive information was gathered from diverse internal and external sources to ensure thorough understanding of requirements, technical approaches, and market context.

2.2.1 Internal Sources

Team Expertise and Academic Resources

- Five team members contributing specialized backgrounds in:
  - Software engineering and system design
  - Database management and architecture
  - Web development (frontend and backend)
  - User experience design and human-computer interaction
  - Project management and documentation

  “‘
- Course materials from System Analysis and Design (CSE-3411)
- Software Engineering principles and methodologies
- Database Management Systems concepts
- Human-Computer Interaction best practices “‘

Faculty and Academic Support

- Regular consultation sessions with course instructor Tasmin Sanjida
- Methodology validation and technical guidance
- Review of previous jobportal system projects for lessons learned
- Access to university library resources

2.2.2 External Sources

Survey Participants

- 57 complete survey responses from diverse demographics
- 91.2
- Distributed through university networks and social media

Questionnaires Surveys

Distributing digital surveys via social media and email to a targeted demographic (fresh graduates, mid-level professionals, and HR managers) to quantitatively measure feature demand (e.g., preference for live classes vs. pre-recorded, automated resume building).

Competitive Benchmarking

Creating a feature-comparison matrix by registering and using competitor services to identify UX/UI flaws and missing functionalities in the current market.

This diverse array of information sources ensured comprehensive understanding of user needs, technical requirements, market opportunities, and implementation challenges, forming a solid foundation for Bridgify development.

2.2.3 Internal Methods

Brainstorming Sessions

Six structured brainstorming sessions conducted over four weeks, addressing:

- Problem identification and project scope definition
- User role identification and stakeholder analysis (jobseeker, job recruiter, admin, assistants)
- Core feature prioritization using MoSCoW method (Must have, Should have, Could have, Won't have)
- Technical stack evaluation and selection criteria
- UI/UX design principles and accessibility requirements
- Comprehensive risk assessment and mitigation strategies
- Timeline estimation and resource allocation

Requirements Workshops

Three intensive workshops where team members adopted different stakeholder perspectives through role-playing exercises to identify comprehensive requirements from each viewpoint, ensuring thorough coverage of all user needs and edge cases.

Technical Research

Distributed research assignments where each team member investigated specific technical domains including frontend frameworks, backend technologies, database systems, real-time communication protocols, payment gateway integration, security best practices, and cloud deployment strategies.

2.2.4 External Sources

The following external methods were used:

Online Survey (Google Forms) A structured questionnaire was distributed to collect quantitative data from users.
‘‘‘

User Interviews Semi-structured interviews were conducted to gain deeper insights into user experiences.

Observation of Existing Platforms Real-world systems were analyzed to understand workflows and usability issues.

User Feedback Analysis Reviews and ratings from different platforms were studied to identify common complaints.

Market Research Trends in job platforms and e-learning systems were examined. ‘‘‘

2.3 Similar Websites

Comprehensive evaluation of Job portal management systems in Bangladesh and international reference platforms:

2.3.1 Bdjobs.com

Overview: The largest and most established online job portal in Bangladesh.
Strengths:

- A massive database of employers and job seekers

- High market trust
- Extensive categorization of corporate and government jobs

Weaknesses:

- The user interface (UI) is outdated and cluttered
- Lacks integrated modern skill-development tracking
- Relies heavily on traditional resume screening rather than skill-based matching

2.3.2 LinkedIn

Overview: A global professional networking platform focusing on B2B connections and career development.

Strengths:

- Excellent for professional branding
- Networking
- Organic headhunting
- Peer-to-peer engagement

Weaknesses:

- Can be intimidating for entry-level candidates
- Premium features are expensive for the local demographic

2.3.3 Coursera

Overview: A leading online learning platform offering courses, certificates, and degree programs from universities and industry partners worldwide.

Strengths:

- High-quality industry-relevant curriculum
- Live mentorship
- Practical project-based learning

Weaknesses:

- Primarily focused on learning rather than direct job placement
- Many valuable certifications require paid enrollment

2.3.4 Indeed

Overview: One of the world's largest job search engines.

Strengths:

- Extensive job database
- Simple search functionality
- Resume hosting

Weaknesses:

- Limited skill verification mechanisms
- Duplicate or outdated job postings
- Minimal learning or career development support

2.3.5 Shikho

Overview: A popular Bangladeshi ed-tech platform focused on digital learning and academic skill development.

Strengths:

- Interactive learning experience
- Localized content
- User-friendly interface
- Strong engagement through quizzes and progress tracking

Weaknesses:

- Primarily designed for education rather than recruitment
- Limited employer integration and job placement functionalities

2.4 Define Desired State

By analyzing the gathered information and evaluating similar websites, a clear gap analysis was performed to define the current limitations and the proposed solution.

2.4.1 Define (Current State)

Currently, the ecosystem in Bangladesh is highly fragmented. A user must navigate multiple disconnected platforms to achieve career growth: using Bdjobs to apply for roles, LinkedIn for networking, and platforms like Ostad or Interactive Cares for upskilling.

Employers struggle to verify the actual skills of applicants because traditional job portals rely on static, easily exaggerated resumes rather than practical assessments.

There is no unified ecosystem that seamlessly connects learning a new skill directly to getting hired for that specific skill.

2.4.2 Desired State

The desired state is to develop an integrated, all-in-one Career Ecosystem.

This platform will bridge the gap between upskilling and recruitment by combining the extensive job-board functionalities of a platform like Bdjobs with the cohort-based skill development models of Ostad/Interactive Cares.

For Job Seekers:

A unified dashboard where they can take courses, build a dynamic profile verified by project completions, and apply for jobs seamlessly.

For Employers:

An AI-driven recruitment dashboard that allows them to hire candidates based on verified skills and project portfolios rather than just academic degrees. The system will automate candidate shortlisting, reducing hiring time and increasing candidate quality.

Chapter 3

System Analysis

3.1 Introduction

This is an essential stage in the lifecycle of CareerBridge Pro, where system analysis reinforces that it satisfies the requirements of students, job seekers and employers. In this stage you study of system requirements, features, and user interactions to direct a consistent, fast and impressive stage.

System analysis for CareerBridge Pro deals mainly with the fine-tuning of its unique features like Smart Profile System, AI Skill Gap Analyzer, Career Path Planner, Course Recommendation System and Job Matching Module to meet user expectations especially those for more up-to-date requirements of students or fresh graduates. The idea is to create an adaptive and intelligent career development system that can customize itself based on user behavior, challenges and the current state of his/her career.

It also finds relationships between multiple modules, the flow of data, and which logic will be driving it all so everything works seamlessly together. Enabling the learning, skill development and job placement features to smoothly interlink into one unified platform. Furthermore, system analysis also assesses how core features like performance tracking, study groups and employer interaction are both technically feasible and user-friendly in order to provide maximum engagement of the users with efficient system operations.

Through this procedure, it identifies relationships between modules, analyzes data flow, and defines integration points for ensuring that systems are communicating smoothly. It integrates learning, skill development and placement into a single ecosystem. Assesses, additionally, the technical feasibility and usability of features such as performance tracking (e.g., scores), description of or access to study groups like clubs or guided/mentored communities and employer interaction with candidates.

In summary, system analysis is the core of CareerBridge Pro, making it a complete and smart platform that helps users develop skills, plan careers and most importantly get employed as efficiently as possible.

3.2 Benchmark

The CareerBridge Pro benchmark will offer a comparative analysis in terms of features, functionality, and performance metrics with other similar systems. It analyzes user experience, personalization, skill analysis, and job matching efficiency. This will guarantee that the system is of industry standard as well as providing a user friendly, integrated and effective career development system.

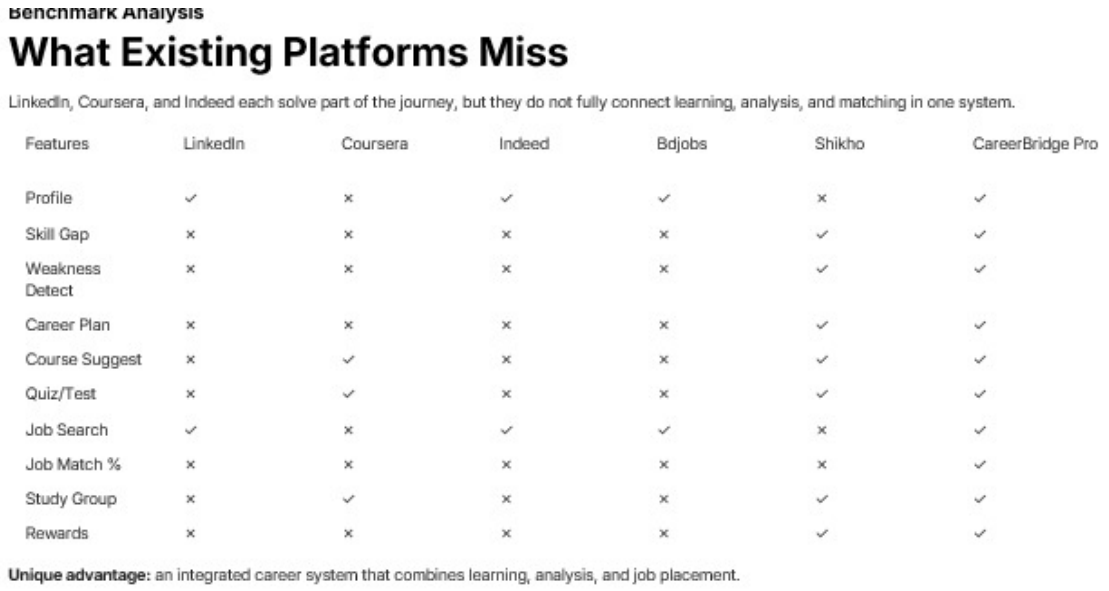


Figure 3.1: Benchmark Analysis of CareerBridge Pro

3.3 Survey

A survey was done to know the needs of users, the challenges, and expectations towards career development, learning, and job searching. This survey was done to obtain real-life information of the potential users and understand the most important issues with existing systems.

The survey assists in knowing how users behave, preferences and requirements which is critical in the design of an effective and user-friendly system. It also helps justify the need to have an integrated platform such as CareerBridge Pro that bridges the learning and skills and employment opportunities.

3.3.1 Survey Methodology

To gauge the potential user opinion, the survey was done on an online questionnaire.

3.3.2 Survey Type

Online questionnaire-based survey.

3.3.3 Target Respondents

- University students
- Fresh graduates
- Job seekers

3.3.4 Sample Size

Approximately 30–50 respondents participated.

3.3.5 Survey Questions Answer

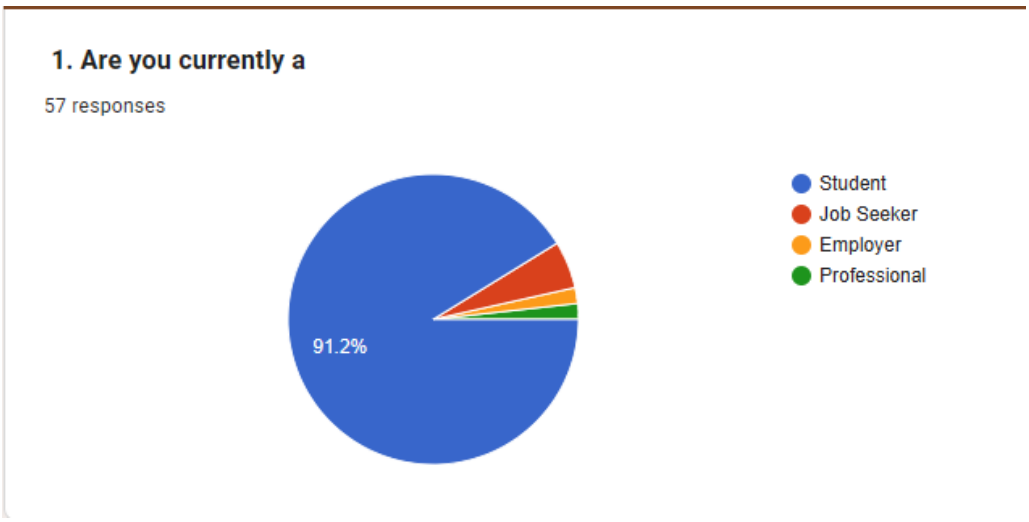


Figure 3.2: Survey Question 01

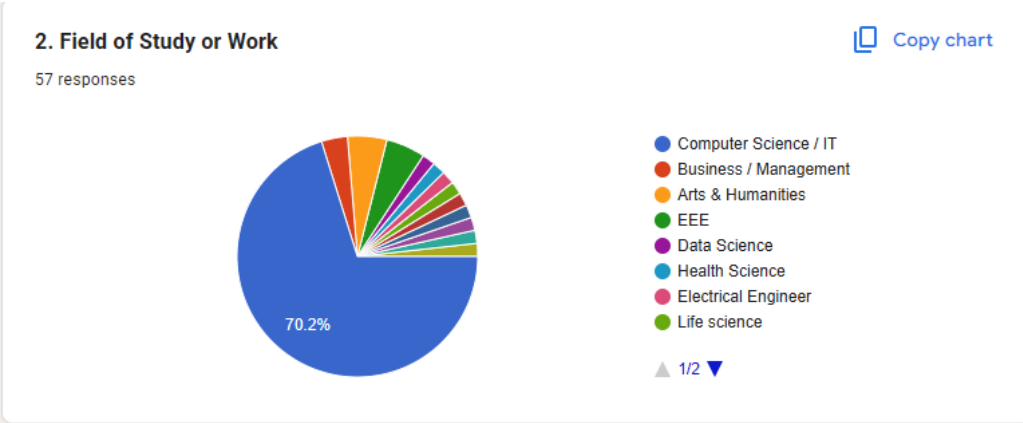


Figure 3.3: Survey Question 02

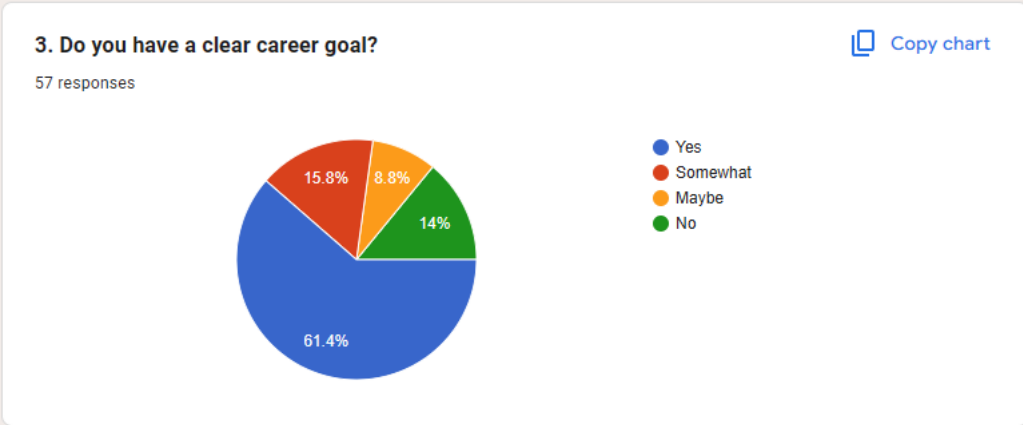


Figure 3.4: Survey Question 03

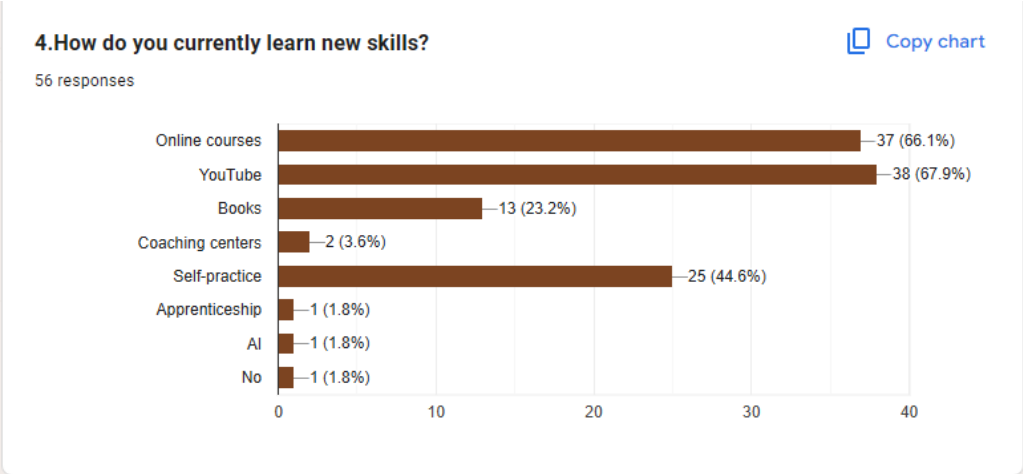


Figure 3.5: Survey Question 04

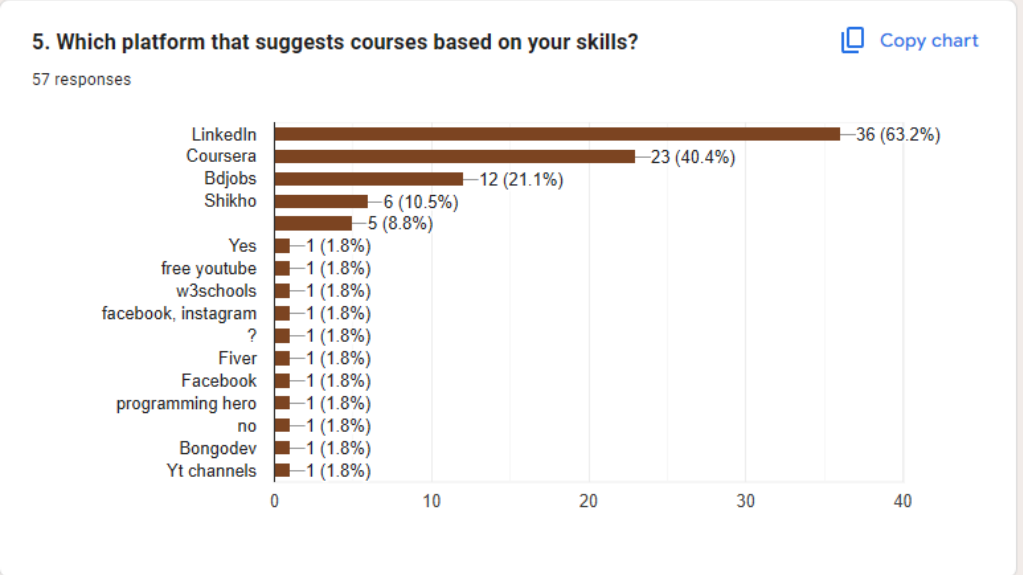


Figure 3.6: Survey Question 05

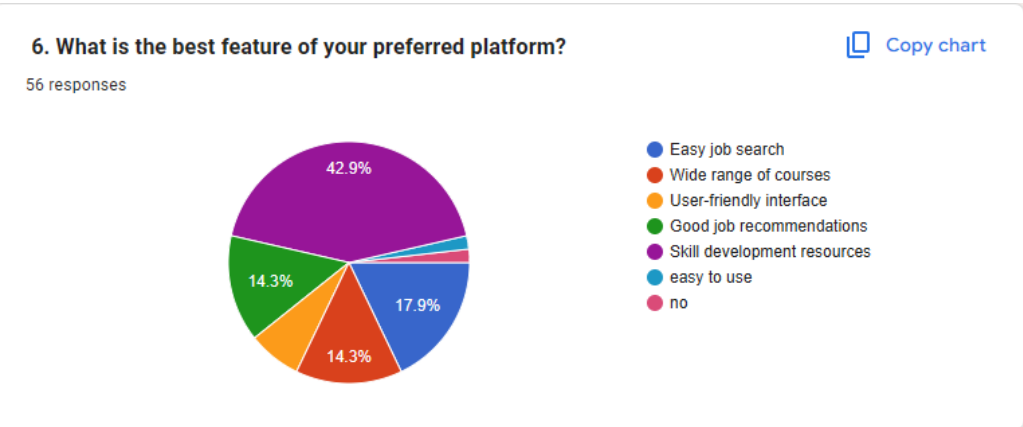


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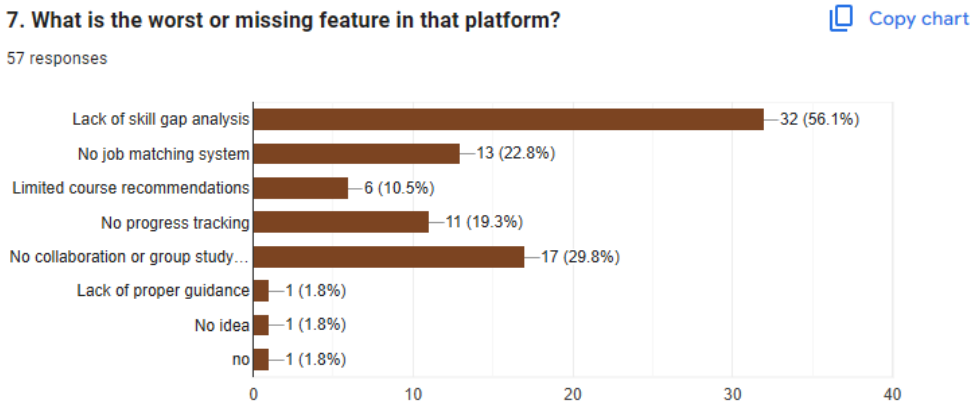


Figure 3.8: Survey Question 07

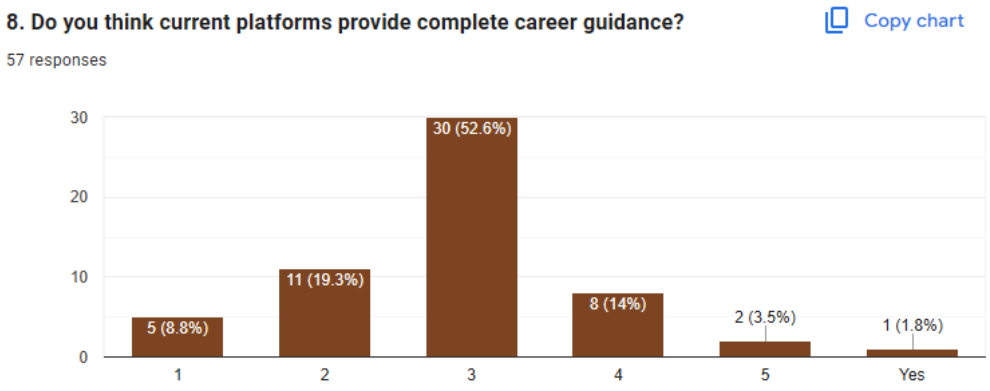


Figure 3.9: Survey Question 08

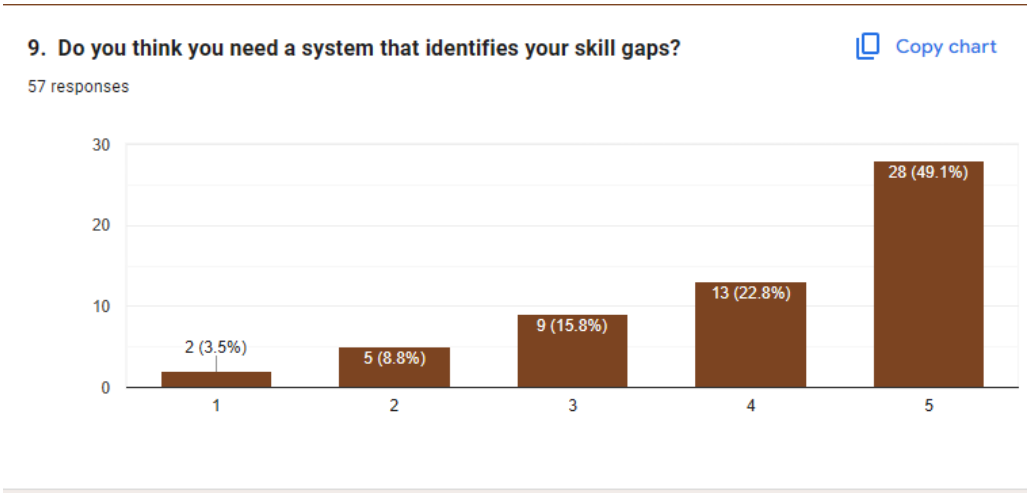


Figure 3.10: Survey Question 09

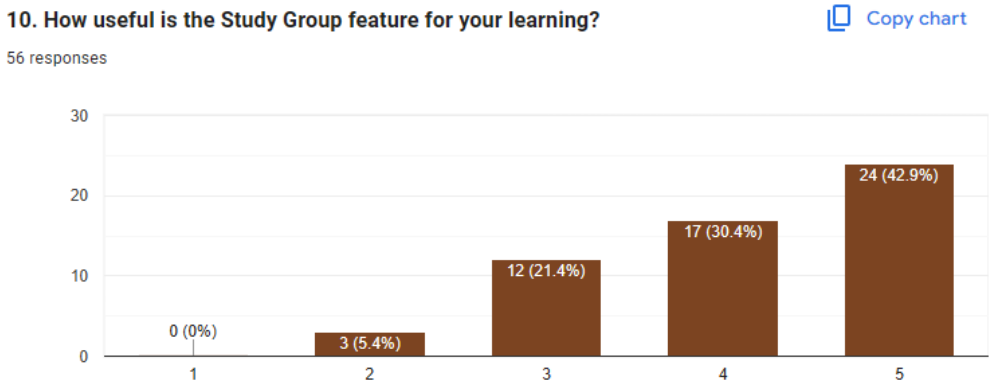


Figure 3.11: Survey Question 10

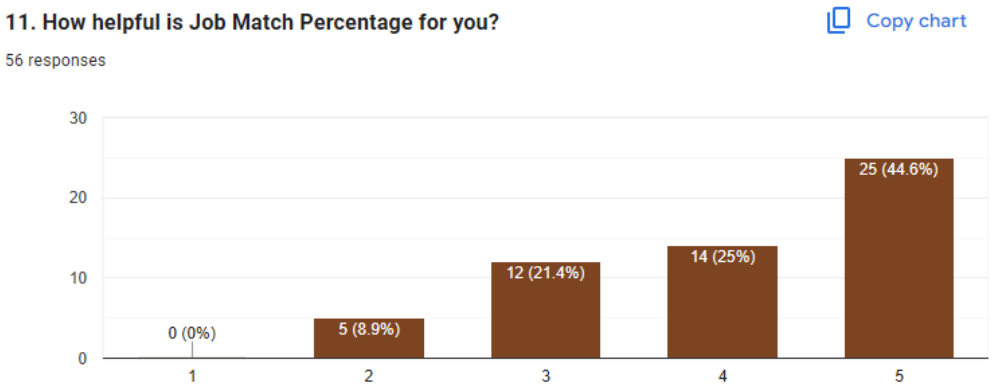


Figure 3.12: Survey Question 11

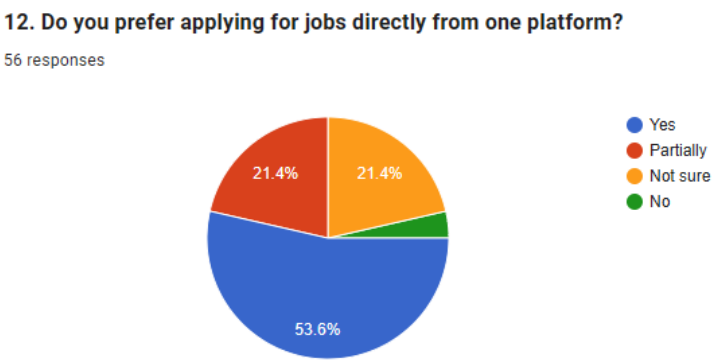


Figure 3.13: Survey Question 12

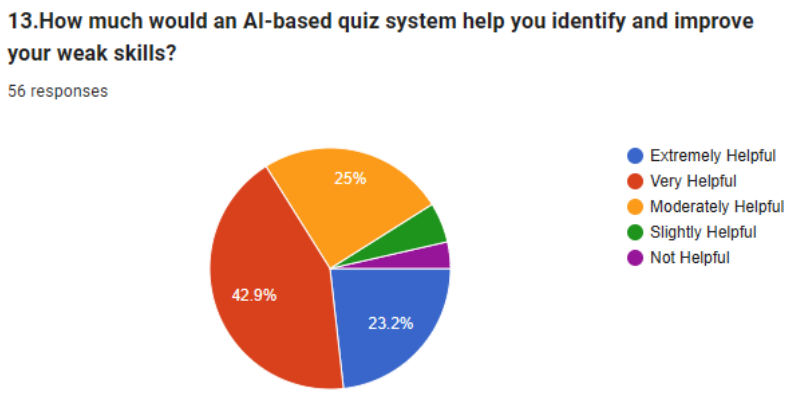


Figure 3.14: Survey Question 13

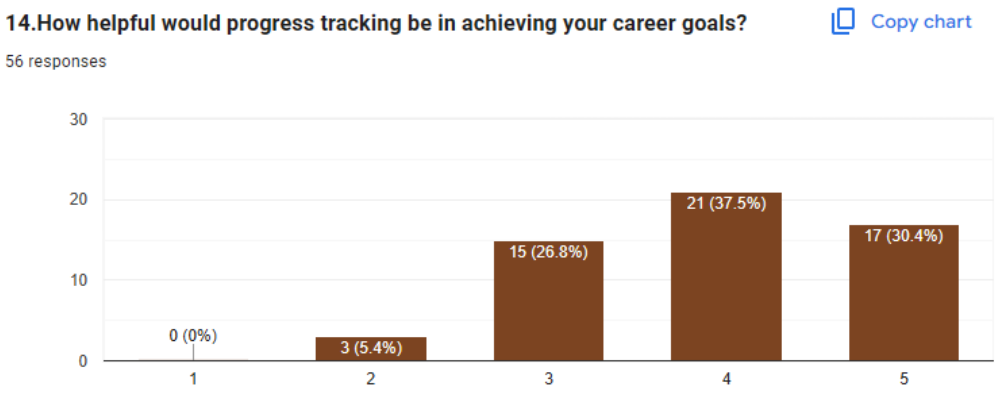


Figure 3.15: Survey Question 14

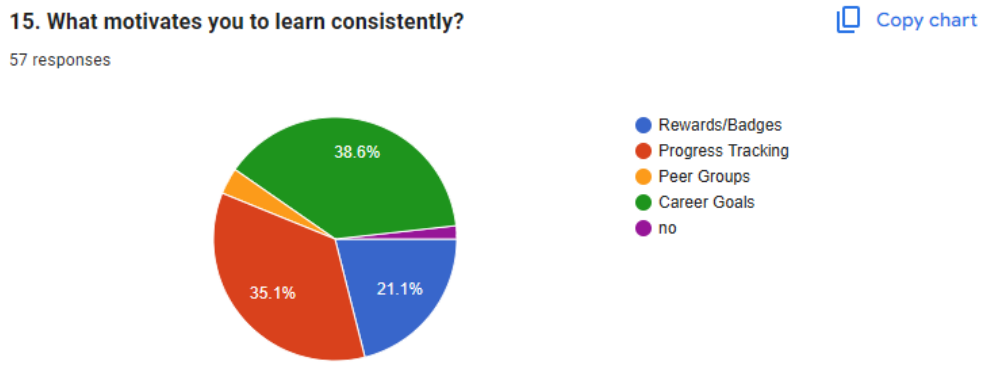


Figure 3.16: Survey Question 16

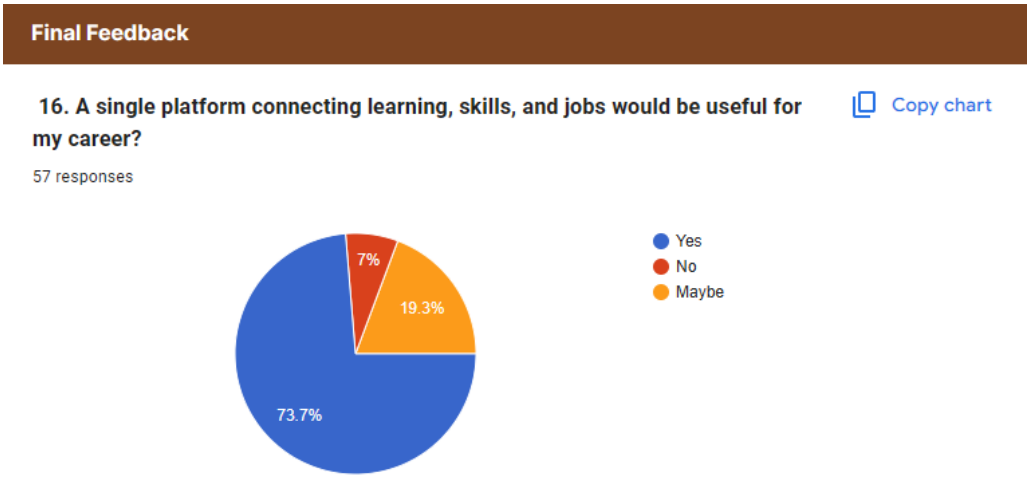


Figure 3.17: Survey Question 16

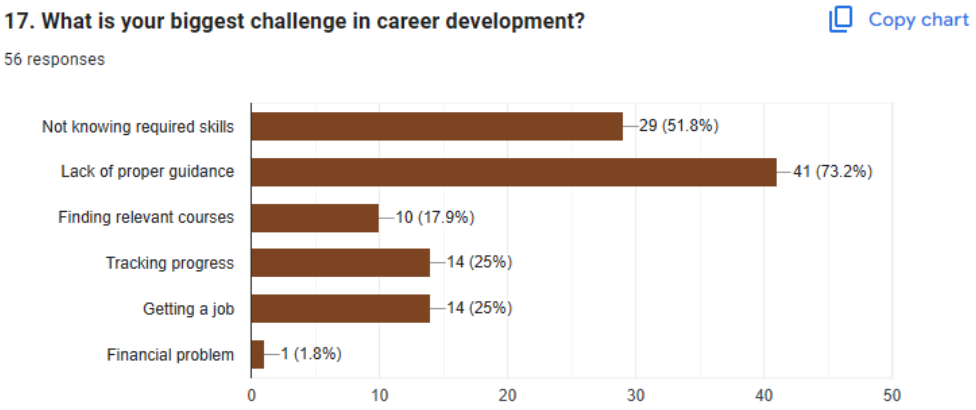


Figure 3.18: Survey Question 17

3.3.6 Some Reviews On Our Survey

Review 01

The concept of integrating various platforms into one is much welcome. An integrated platform that contains pertinent skills, ratings, and successful rates can assist the users in making the right decisions and also save some time in selecting the appropriate career path.

Review 02

The users proposed that appropriate guidelines by industry experts should be included. Professional guidance and practical experience might assist the users to know more about the requirements of the job and prepare themselves in careers.

Review 03

The respondents stressed the value of such advanced features as the analysis of skill gap and mock interview. These capabilities would assist users in determining their areas of weakness and to enhance job preparation.

3.4 Gap Analysis

These are the existing limitations of the current system (Current Platforms).

1. **Fragmented System Structure:**
    - Separate platforms are offered to learn, develop skills, and find a job, which is inefficient and time-consuming.

2. **There is No Skill Gap Analysis:**
    - Current platforms fail to identify weak or missing skills of users required in particular jobs.
3. **Absence of Career Path Advising:**
    - There is no clear roadmap to the goals of the career that can be given to users.
4. **Inefficient Job Matching:**
    - Job recommendations are largely key-word based and do not take actual skill compatibility into account.
5. **Limited Personalization:**
    - The majority of systems do not offer individual recommendations, but general ones according to user profiles.
6. **No Integrated Learning Support:**
    - Learning platforms exist independently, and are not directly linked with job requirements.
7. **Weak Progress Tracking:**
    - Current systems fail to effectively monitor user learning and improvement of skills with time.
8. **None of the Job Match Percentage Features:**
    - Platforms lack a clear representation (percentage) of the fit of a user to a job.
9. **Little Conducive to Fresh Graduates:**
    - Novices and learners usually lack access to an appropriate guidance or entry-level career support.
10. **Absence of Unified Career System:**
    - There is no platform where one can learn, analyze skills and get a job at a single location.

3.5 Feature List Fixation

Based on the system analysis, survey results, and gap analysis, the following features are finalized for CareerBridge Pro.

Core Features

- User Registration Login
- Smart Profile System
- Job Search Application
- Employer Panel

AI-Based Features

- AI Skill Gap Analyzer
- Career Path Planner
- Job Match Percentage

Learning Features

- Course Recommendation
- Online Quiz System
- Study Group System

Support Engagement Features

- Performance Tracking
- Notifications Rewards
- Mock Interview System

Chapter 4

Functional Requirements

CareerBridge Pro is designed as a comprehensive career development platform that integrates learning, skill analysis, and job placement into a unified system. The system aims to guide users step by step from identifying their current skill level to achieving their desired career goals.

Foundation for Career Growth

4.1 Smart Profile System

The Smart Profile System serves as the core component of the platform, allowing users to create and manage a complete professional profile. Users can add their education, skills, achievements, and career goals in one centralized place. This structured profile acts as a foundation for other modules such as skill gap analysis, personalized recommendations, and job matching. It ensures that all system decisions are based on accurate and up-to-date user data.

Inputs

- User personal information
- Education details
- Skills, achievements, career goals

Outputs

- Updated user profile
- Stored profile data

Interactions

- Provides data to AI Skill Gap Analyzer
- Used by Job Matching System
- Supports Course Recommendation System

4.2 AI Skill Gap Analyzer

This feature evaluates the user's current skills and compares them with industry standards and job-specific requirements. It identifies missing and weak skills that need improvement. By providing detailed insights and highlighting areas of improvement, this feature enables users to focus on relevant skills rather than learning randomly, thus making the learning process more efficient and goal-oriented.

Inputs

- User skills
- Target job role
- Industry requirements data

Outputs

- List of missing skills
- Weak skill identification
- Improvement suggestions

Interactions

- Receives data from Smart Profile System
- Sends results to Career Path Planner
- Supports Course Recommendation System

4.3 Career Path Planner

The Career Path Planner provides a structured and personalized roadmap based on the user's selected career goal. It outlines the necessary skills, recommended learning steps required to reach the desired role. This feature eliminates confusion by guiding users through a clear sequence of actions, helping them move forward in a systematic and organized manner.

Inputs

- Target career
- Skill gap analysis data

Outputs

- Career roadmap
- Required skill list
- Learning steps

Interactions

- Uses data from AI Skill Gap Analyzer
- Supports Course Recommendation System
- Guides Mock Interview preparation

4.4 Performance Tracking

The Performance Tracking module continuously monitors user progress by analyzing quiz results, learning activities, completed courses and overall activity within the platform. It visually represents improvement and performance levels. This allows users to clearly understand their improvement over time and stay motivated by observing measurable growth.

Inputs

- Quiz results
- Course completion data

- User activity

Outputs

- Progress reports
- Performance analytics
- Improvement status

Interactions

- Receives data from Online Quiz System
- Tracks Course Recommendation progress
- Supports Job Matching readiness

Make Improvement Social and Personalized

4.5 Study Group System

The Study Group System connects users with similar goals and skill levels. It allows members to communicate, share resources, discuss problems in a group environment and solve problems together. This social aspect of learning increases engagement, encourages peer support, and helps users learn more effectively through interaction and teamwork.

Inputs

- User interests
- Skill levels
- Learning goals

Outputs

- Group suggestions
- Group activity updates

Interactions

- Uses Smart Profile data
- Supports Online Quiz discussion
- Enhances learning collaboration

4.6 Course Recommendation System

The Course Recommendation System provides personalized learning suggestions based on the user's profile, identified skill gaps, and career objectives. It filters and recommends relevant courses, tutorials, and learning materials. It ensures that users focus only on necessary topics, saving time and improving learning efficiency.

Inputs

- Skill gap data
- User profile
- Career goals

Outputs

- Recommended courses
- Learning resources list

Interactions

- Receives data from AI Skill Gap Analyzer
- Supports Career Path Planner
- Helps Online Quiz System

4.7 Online Quiz System

The Online Quiz System offers skill-based assessments to evaluate user knowledge in different subject areas. It generates quizzes based on the user's learning stage and target skills. It helps identify weak areas and measure improvement. Regular quizzes ensure continuous learning and better preparation for real-world challenges.

Inputs

- Quiz answers
- Selected skill

Outputs

- Quiz scores
- Performance feedback

Interactions

- Sends results to Performance Tracking
- Uses Course Recommendation topics
- Supports Skill Gap re-evaluation

4.8 Mock Interview System

The Mock Interview System simulates real-life interview scenarios to prepare users for job recruitment processes. It may include technical questions, behavioral questions, and timed responses. Users can practice answering questions and improve communication skills. It increases confidence and readiness for actual job applications.

Inputs

- User responses
- Target job role

Outputs

- Interview feedback
- Performance score

Interactions

- Uses Career Path Planner data
- Supports Job Matching readiness

- Improves user confidence

Move from Learning to Placement

4.9 Job Matching System

The Job Matching System analyzes user profiles, skills, and performance data to recommend suitable job opportunities. It provides a match percentage to indicate how well the user fits a job. This intelligent matching system reduces irrelevant applications and helps users focus on opportunities where they have a higher chance of success.

Inputs

- User profile
- Skills and performance data
- Job database

Outputs

- Job recommendations
- Match percentage

Interactions

- Uses Smart Profile System
- Uses AI Skill Gap Analyzer data
- Connects with Employer Panel

4.10 Application System

The Application System allows users to apply for jobs directly through the platform without switching systems. It simplifies the application process by enabling quick submissions using stored profile data. The process is simple and user-friendly. This seamless integration reduces the need for external platforms and ensures a smooth transition from skill development to job application.

Inputs

- Selected job
- User profile data

Outputs

- Job application submission
- Confirmation message

Interactions

- Sends applications to Employer Panel
- Uses Job Matching results
- Triggers Notifications

4.11 Employer Panel

The Employer Panel allows companies to post job vacancies and manage applicants efficiently. Employers can filter candidates based on skills and qualifications. It enhances the recruitment process by allowing employers to efficiently identify and select the most suitable candidates from a pool of applicants.

Inputs

- Job details
- Candidate filters

Outputs

- Job postings
- Candidate shortlist

Interactions

- Receives applications from Application System
- Connects with Job Matching System

4.12 Notifications Rewards System

The Notifications Rewards System keeps users informed and motivated throughout their journey. It sends alerts for important activities such as job deadlines, study reminders, and updates. Additionally, it incorporates a reward mechanism including badges, points, and achievements to encourage consistent engagement and continuous improvement.

Inputs

- User activity
- System events jobs, deadlines, updates

Outputs

- Notifications
- Rewards badges, points

Interactions

- Connected with all modules
- Sends alerts to users
- Supports user engagement

Chapter 5

External & Non-Functional Requirements

5.1 External & Non-Functional Requirements

5.1.1 User Interfaces

The system provides a modern, responsive, and user-friendly interface for both web and mobile users.

- **Frontend Technology:** React.js with Tailwind CSS (or similar modern UI framework)
- **Design Style:** Clean dashboard-based UI with role-specific views (Student, Admin, User,Recruiter)
- **Navigation System:**
  - Sidebar navigation for dashboard modules (Skill Gap Analysis, Career Path, Job Matching, Profile)
  - Top navigation bar for quick actions (notifications, profile, logout)
- **Key Pages:**
  - Landing Page (marketing + feature overview)
  - Authentication Pages (Login/Register via Firebase Auth)
  - Dashboard (role-based access)
  - Skill Gap Analysis Module
  - Career Path Recommendation Page
  - Job Matching Interface

  - Performance & Reward Dashboard

- **Prototype Alignment:**
  - Desktop UI includes full feature modules (11 major screens)
  - Mobile UI is simplified (7 core screens: Landing, Skill Gap, Dashboard, Performance, Mock Tests, Profile, Employee View)

5.1.2 Software Interfaces

The system integrates with multiple external and internal services to ensure full functionality.

- **Frontend–Backend Communication:** REST API using Node.js + Express.js
- **Data Format:** JSON-based data exchange
- **Database:** MongoDB for storing user profiles, career data, job matches, and performance records
- **Authentication Service:** Firebase Authentication (Email/Password + Google Login)
- **AI Integration:** AI-based recommendation engine for skill gap analysis and career suggestions
- **External APIs (optional expansion):**
  - Job listing APIs for real-time job matching
  - Email service for notifications and alerts

5.1.3 Performance Requirements

The system is designed for fast response, scalability, and smooth user experience.

- **Response Time:**
  - Dashboard load time $\leq$ 2–3 seconds under normal load
  - API response time $\leq$ 500ms for standard requests
- **Scalability:**
  - Supports 1,000+ concurrent active users (initial target)
  - Modular architecture allows horizontal scaling of backend services

- **Reliability:**
  - 99% system uptime target
  - Graceful error handling for API failures and network issues
- **Optimization:**
  - Lazy loading for React components
  - Pagination for large datasets (jobs, users, courses)

5.1.4 Security Requirements

Security is a critical component due to user data and authentication-sensitive operations.

- **Authentication & Authorization:**
  - Firebase Authentication with secure token-based login
  - Role-based access control (Admin, User, Counselor)
- **Data Protection:**
  - Passwords are never stored in plain text
  - Secure API communication using HTTPS
  - Environment variables for sensitive keys
- **Access Control:**
  - Protected routes in frontend (React Router guards)
  - Backend middleware validation for admin-only routes
- **Security Enhancements:**
  - Input validation to prevent SQL/NoSQL injection
  - Rate limiting on APIs to prevent abuse

5.1.5 Maintainability

The system is designed to be easy to maintain, extend, and update.

- **Modular Architecture:**
  - Separate frontend components (reusable UI components)
  - Backend structured using MVC pattern (Controllers, Routes, Models)

- **Code Quality:**
  - Clean, readable, and well-documented codebase
  - Consistent naming conventions and folder structure
- **Scalability for Features:**
  - Easy integration of new modules (AI chatbot, analytics tools)
- **Deployment & Updates:**
  - Frontend deployable via Vercel/Netlify
  - Backend deployable via Render/Heroku/VPS
  - CI/CD pipeline support (GitHub Actions optional)
- **Testing & Debugging:**
  - API testing via Postman
  - Component-level testing in React (future enhancement)

Chapter 6

Feasibility & Project Management Analysis

6.1 Feasibility Analysis

6.1.1 Technical Feasibility

Let's break down how practical CareerBridge Pro really is.

On the technical side, the team's set to use proven tools: React and Next.js for the front, Node.js with Express.js for the backend, MongoDB for data storage, and Firebase/Auth to keep things secure. Nothing unusual here—and that's a good thing.

The tricky part? Developing that AI Skill Gap Analyzer and Job Matching System. Those features need a solid database and smart AI logic to work right and actually mean something to users.

6.1.2 Operational & Market Feasibility

Operationally and in the market, there's a real need. Other platforms—think LinkedIn, Coursera, Bdjobs—only focus on bits and pieces. No one puts everything together: learning, skills, job matching.

Students and job seekers keep saying the same things—they're lost, struggling to figure out what to do next, and want advice and mentorship.

So, by focusing on students and recent grads and offering what they ask for—course suggestions, group study tools, mock interviews—CareerBridge Pro fits right in. Plus, there's room to grow, since EdTech and job markets in Bangladesh are taking off.

6.2 SWOT Analysis

Strengths	**Weaknesses**
• One-stop shop—it brings learning, skills, and jobs all together. • AI-powered features such as Skill Gap Analyzer and Job Match Percentage. • Personalized career roadmaps based on user goals and interests.	• Requires accurate and complete user data for optimal AI performance. • High complexity in backend and AI system development. • Dependence on continuous data updates for accuracy and relevance.
Opportunities	**Threats**
• High demand for integrated career development platforms. • Strong partnership opportunities with universities and training centers. • Rapid growth of EdTech and job markets in Bangladesh.	• Strong competition from LinkedIn, Coursera, and Bdjobs. • User trust and adoption barriers for new platforms. • Fast-changing technologies requiring continuous updates.

Table 6.1: SWOT Analysis of CareerBridge Pro

6.3 Strategic Recommendations

6.3.1 Show Off What Makes Us Unique

Let's spotlight our Skill Gap Detection and Job Match percentages—these are the real game-changers, not just the usual features you find everywhere.

6.3.2 Keep Things Simple and Personal

The platform should feel easy to use, not cluttered. Give people recommendations that matter, not just a bunch of generic tips.

6.3.3 Focus on Who Needs Us Most

Target students and fresh graduates first. They're searching for direction and are the ones who'll really benefit from what we do.

6.3.4 Form Real Partnerships

Work closely with universities, training centers, and businesses. These connections help us reach more users and boost our growth.

6.3.5 Continuous Improvement

Pay attention to user feedback. Listen, update, and fine-tune the platform so it gets more accurate and useful every time.

6.3.6 Gradual Scaling

No need to rush. Launch new features gradually so we can keep up with demand and reach more people along the way.

Chapter 7

Diagrams

7.1 DFD Diagram

7.2 Use Case Diagram

7.3 Use Case Description

7.3.1 Use Case ID: UC-1

7.4 Activity Diagram

7.5 Sequence Diagram

7.6 Class Diagram

7.7 State Diagram

7.8 Swimlane Diagram

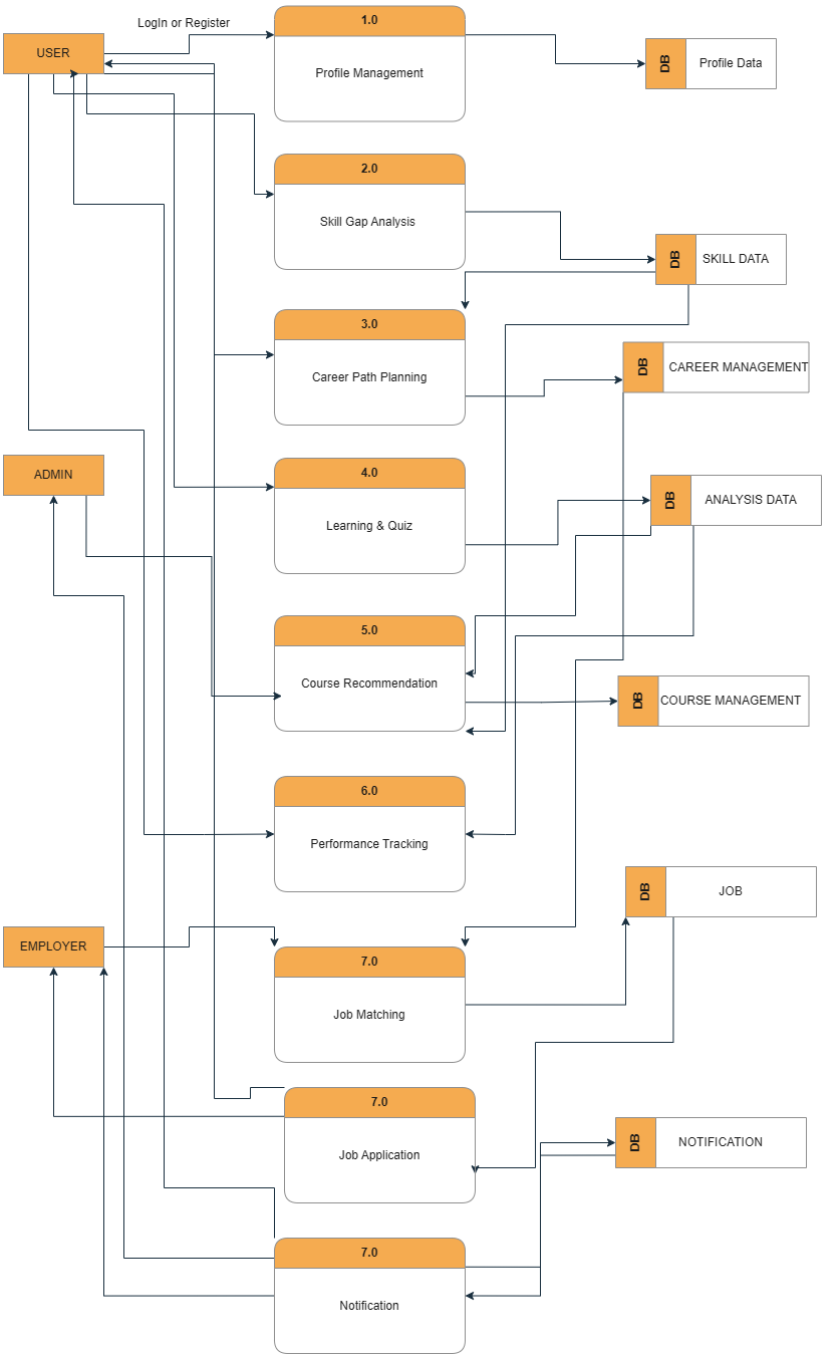


Figure 7.1: CareerBridge Pro DFD Diagram

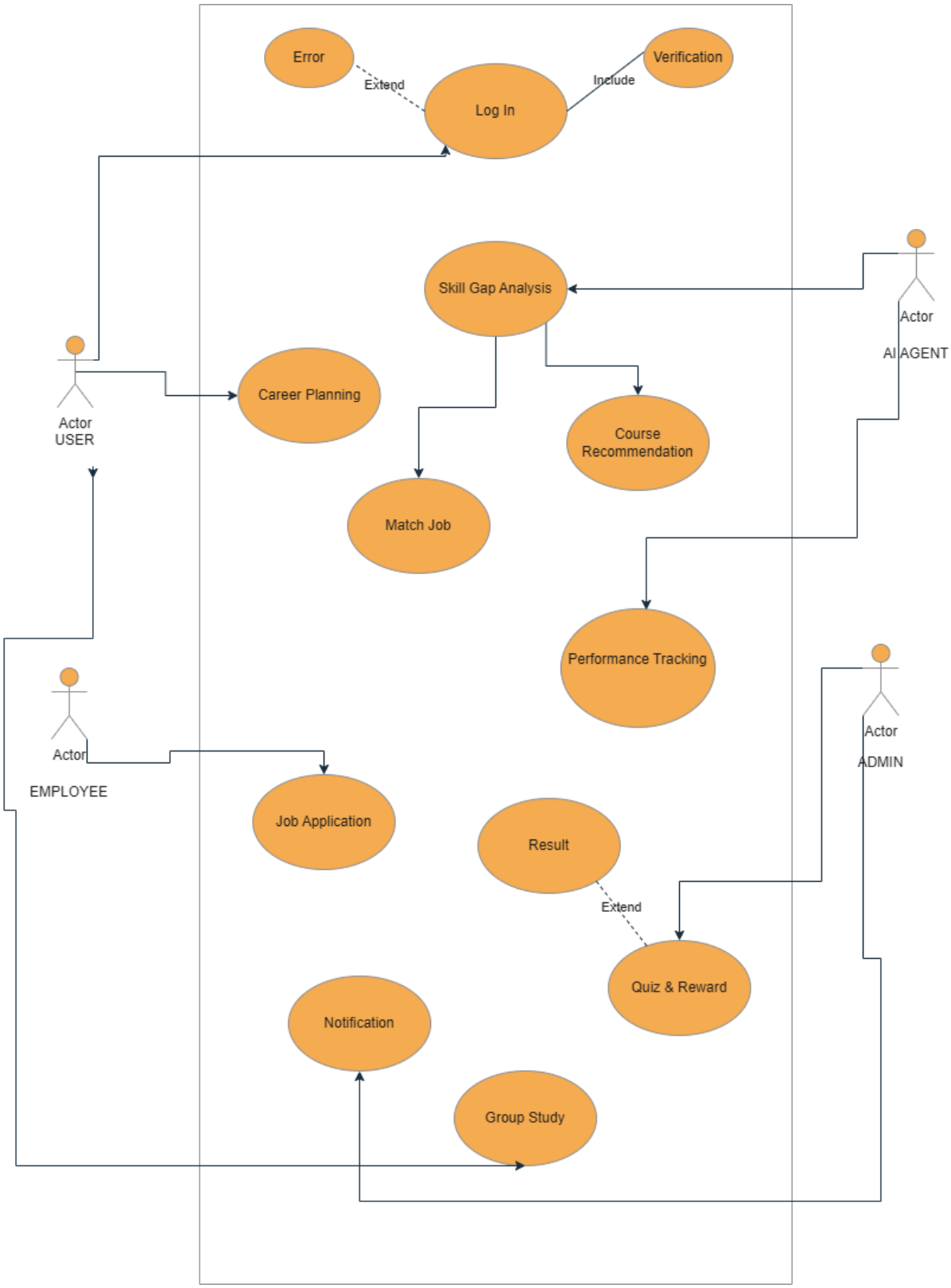


Figure 7.2: CareerBridge Pro Use Case Diagram

Use Case ID	UC-1
Use Case Name	Job Application
Description	Allows the user to apply for specific job openings that match their skills and career goals.
Stakeholders	• User (Job Seeker) • Employer (Company) • AI Agent (for match tracking) • Admin
Primary Actor	User
Pre-condition	The user must be registered, logged into the system, and have a completed profile, while the skill assessment quiz is optional for better job recommendations.
Success Scenario	1. User logs into the system. 2. User views AI-generated matched jobs. 3. User selects a job and reviews the details. 4. User clicks the ”Apply” button. 5. The system submits the application. 6. The system updates the application status on the user’s dashboard.
Alternative Scenarios	• If the user’s profile is missing required skills, the system prompts them to update their profile before applying. • If the user has already applied for the same job, the system displays a message indicating that duplicate applications are not allowed. • If a server error occurs during submission, the system alerts the user and asks them to try again later.
Post-condition	The application is successfully submitted to the employer, and the user receives a confirmation notification.

Table 7.1: Use Case Specification for Job Application

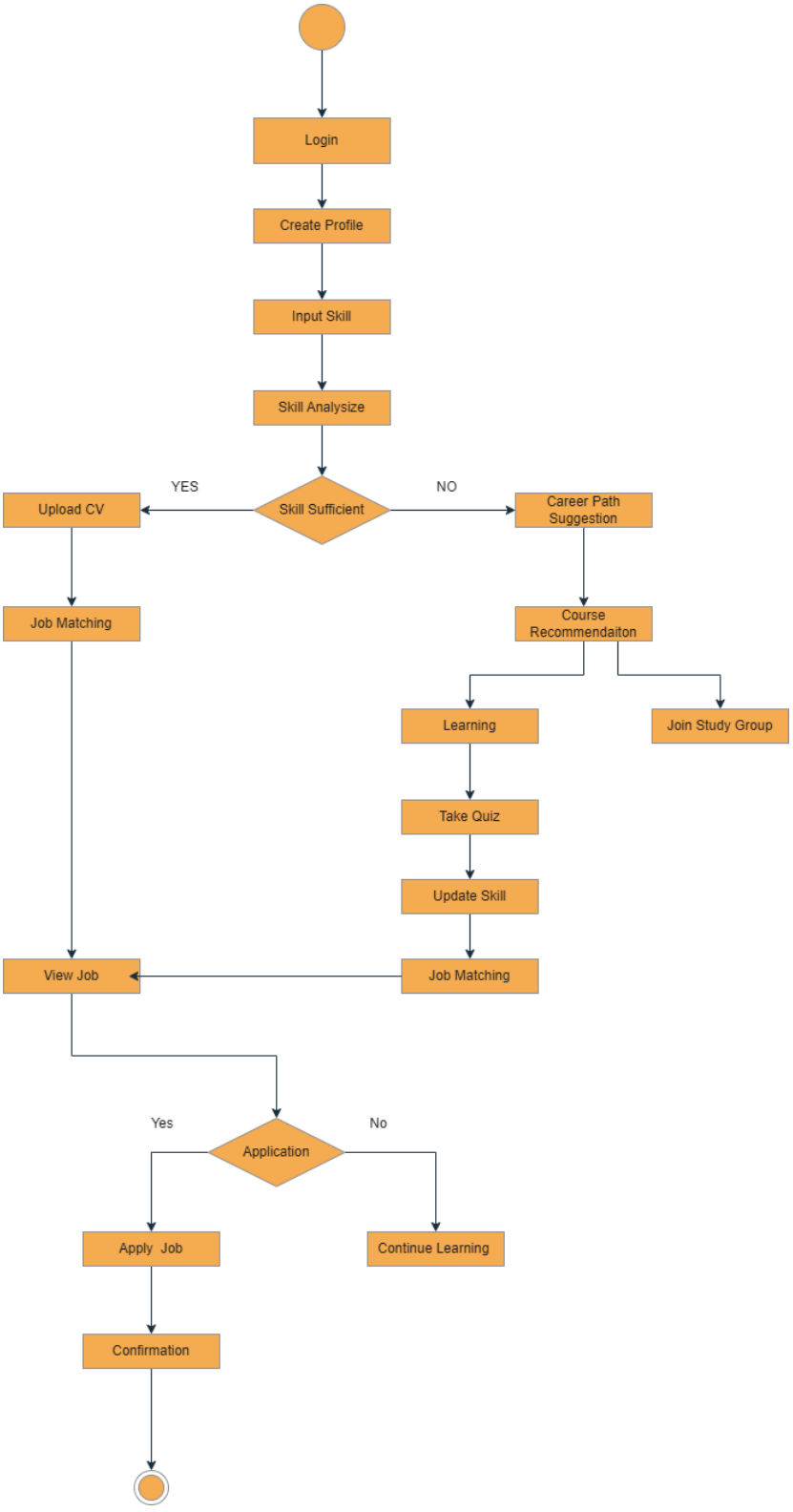


Figure 7.3: CareerBridge Pro Activity Diagram

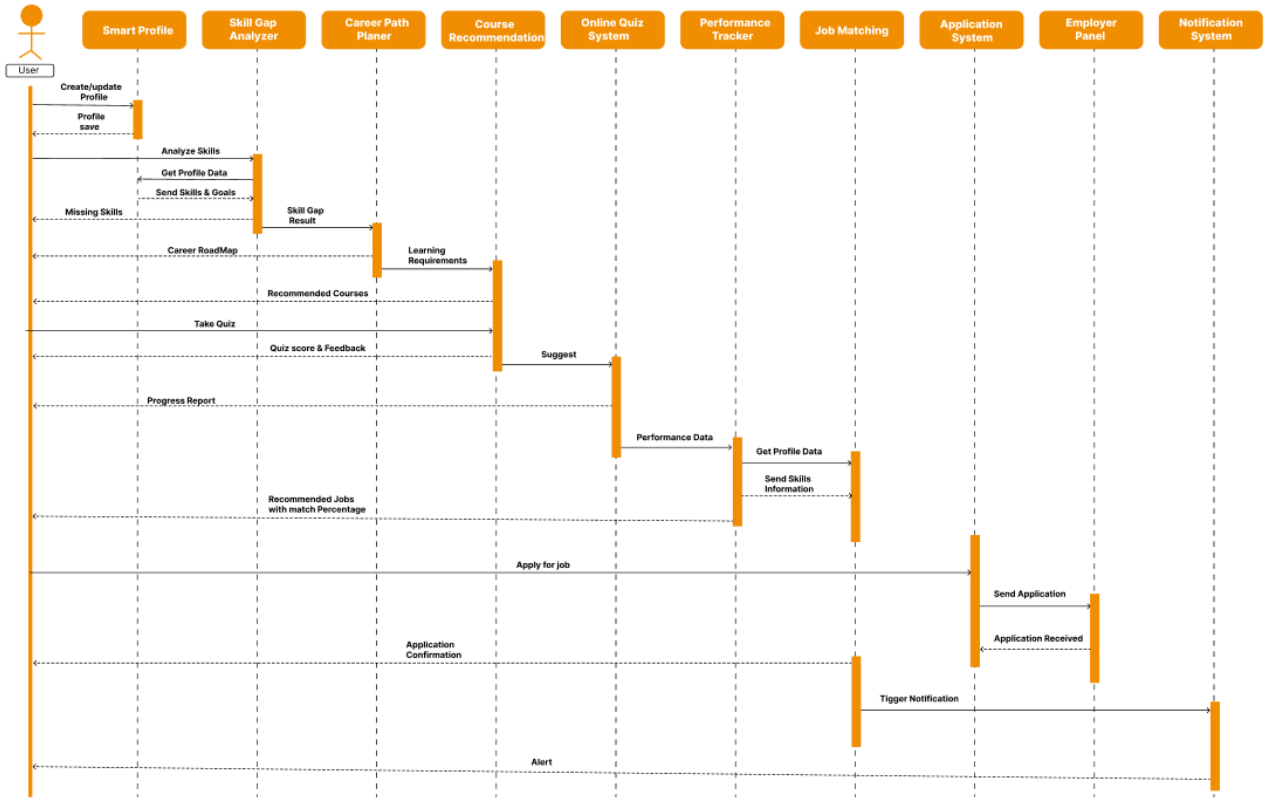


Figure 7.4: CareerBridge Pro Sequence Diagram

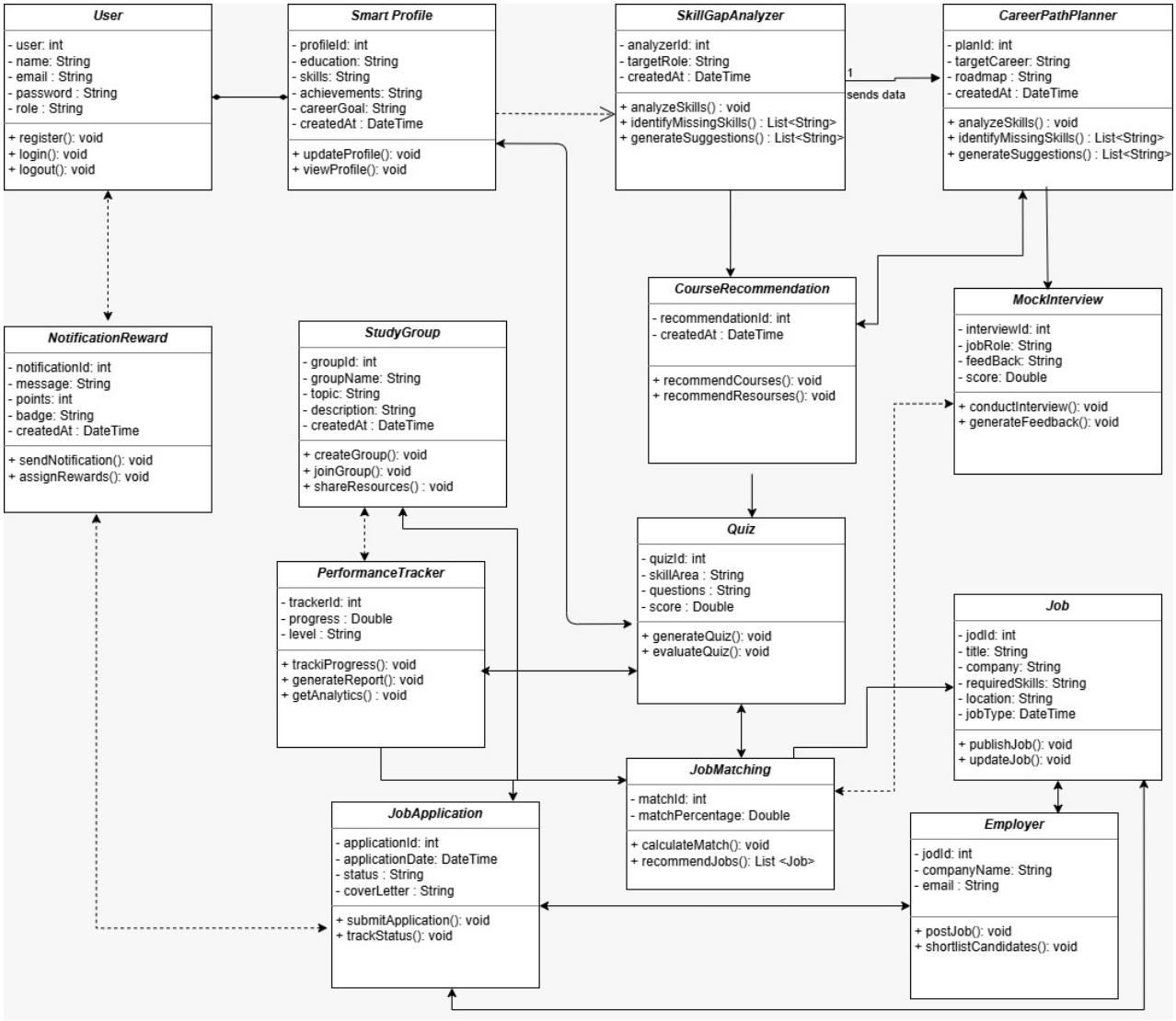


Figure 7.5: CareerBridge Pro Class Diagram



Figure 7.6: CareerBridge Pro State Diagram

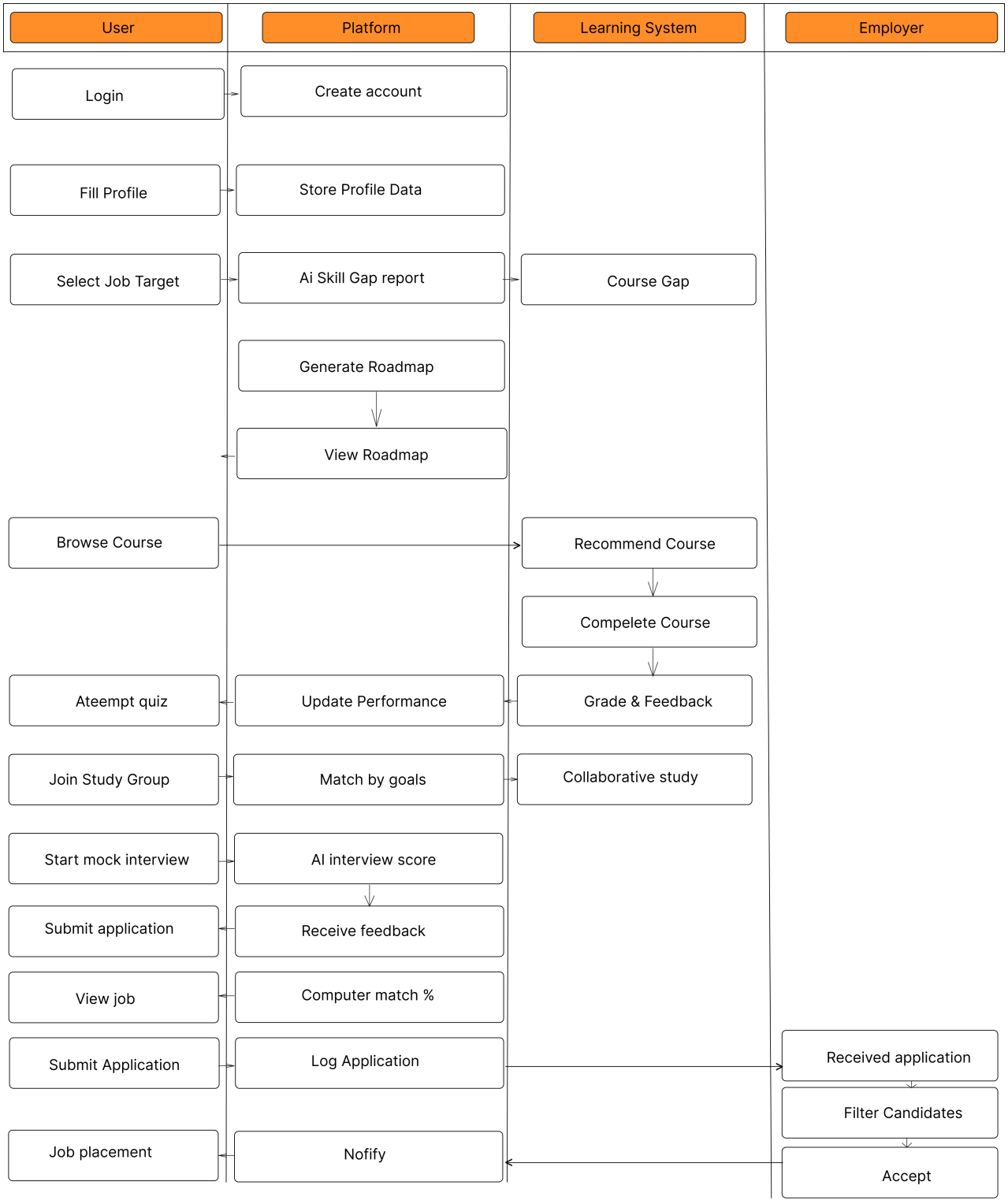


Figure 7.7: CareerBridge Pro Swimline Diagram

Chapter 8

Prototype Design

8.1 Introduction

This chapter outlines the prototype and UI design process for Project CareerBridgePro, an interactive digital learning platform aimed at enhancing academic engagement through gamification and AI-based learning for students from Class 6 to graduate level. The goal of the UI design is to create an intuitive, competitive, and user-friendly experience that aligns with the platform's educational and entertainment goals. The chapter details the tools used for prototyping, the key elements of the UI design, and the overall sitemap.

8.2 Prototyping Tools Used

To ensure an effective and visually appealing prototype, the following tools were utilized:

- **Figma:** Used for designing interactive wireframes and high-fidelity prototypes. Figma enabled real-time collaboration for design feedback and iterative improvements.
- **Canva:** Assisted in creating visual assets like banners, icons, and promotional graphics.

8.3 UI Design

The UI design of Project CareerBridgePro emphasizes competitiveness, clarity, and motivation to engage young learners in an educational environment enriched with game-like experiences.

8.3.1 Rules of UI Design

Following standard design principles, the platform adopts Mandel's Golden Rules, categorized as:

Place Users in Control

1. Use modes judiciously
2. Allow users to use either keyboard or mouse
3. Allow users to change focus
4. Display descriptive messages and text
5. Provide immediate and reversible actions and feedback
6. Provide meaningful paths and exits
7. Accommodate users with different skill levels
8. Make the user interface transparent
9. Allow users to customize the interface
10. Allow users to directly manipulate interface objects

Reduce Users' Memory Load

1. Relieve short-term memory
2. Rely on recognition, not recall
3. Provide visual cues
4. Provide defaults, undo, and redo
5. Provide interface shortcuts
6. Promote object-action syntax
7. Use real-world metaphors
8. Use progressive disclosure
9. Promote visual clarity

Make the Interface Consistent

1. Sustain user context
2. Maintain consistency within and across products
3. Keep interaction results the same
4. Provide aesthetic appeal and integrity
5. Encourage exploration

8.4 Color Palette

- **Primary Colors:** Orange and white invoke excitement and focus.
- **Secondary Colors:** Warm oranges balance engagement and approachability.
- **Accent Colors:** Orange and white are used for important actions such as Apply Now, Start Quiz, View Career Roadmap, and Update Profile.

8.5 Typography

- **Headings:** Poppins font for clear hierarchy.
- **Body Text:** Inter or Roboto for readability.
- **Accessibility:** Proper spacing, contrast, and font sizing for all devices.

8.6 Layout

- Sidebar navigation includes Dashboard, Profile, Skill Analyzer, Career Path, Courses, Jobs, Performance, Mock Interview, Rewards, and Notifications.
- Card-based UI used for jobs, courses, and analytics.
- Fully responsive design for desktop, tablet, and mobile devices.

8.7 Key Features

- AI Skill Gap Analyzer
- Career Path Planner

- Course Recommendations
- Job Matching System
- Quizzes and Challenges
- Mock Interview System
- Performance Tracking
- Rewards and Achievements
- Study Groups and Community
- AI Assistant

8.8 Sitemap

The sitemap defines the structure of CareerBridgePro, organizing learning, career planning, and job-related modules into a user-friendly navigation system that ensures easy access to all features.

8.9 Web Prototype Screens (Desktop)

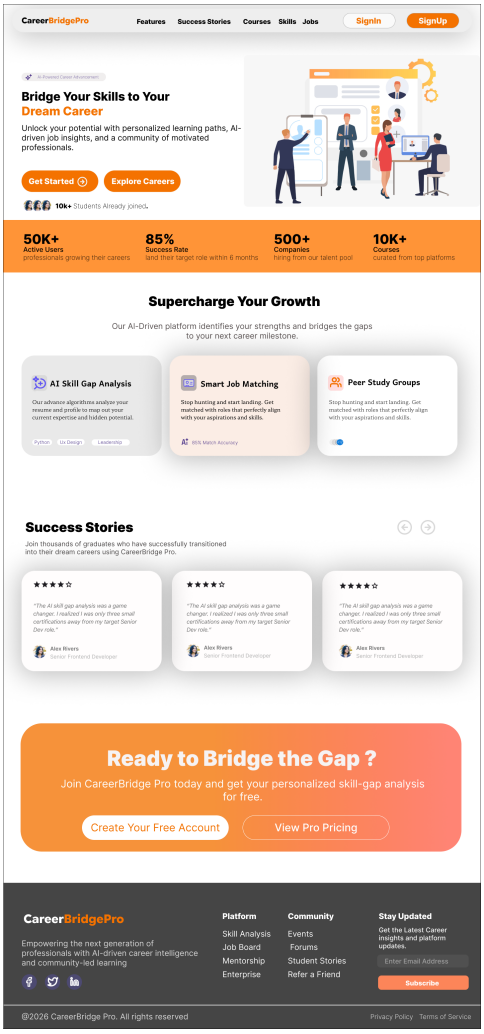


Figure 8.1: Desktop Landing Page

Landing Page

- Welcome message introducing CareerBridge Pro and its mission to bridge the gap between learning, skill development, and employment opportunities.
- Options to Sign Up, Log In, and explore the platform.
- Overview of key features such as AI Skill Gap Analysis, Career Path Planning, Course Recommendations, Job Matching, and Mock Interviews.
- Featured success stories, user testimonials, and career growth statistics.
- Quick access to recommended jobs, learning resources, and platform highlights.

8.9.1 Skill Gap Analysis

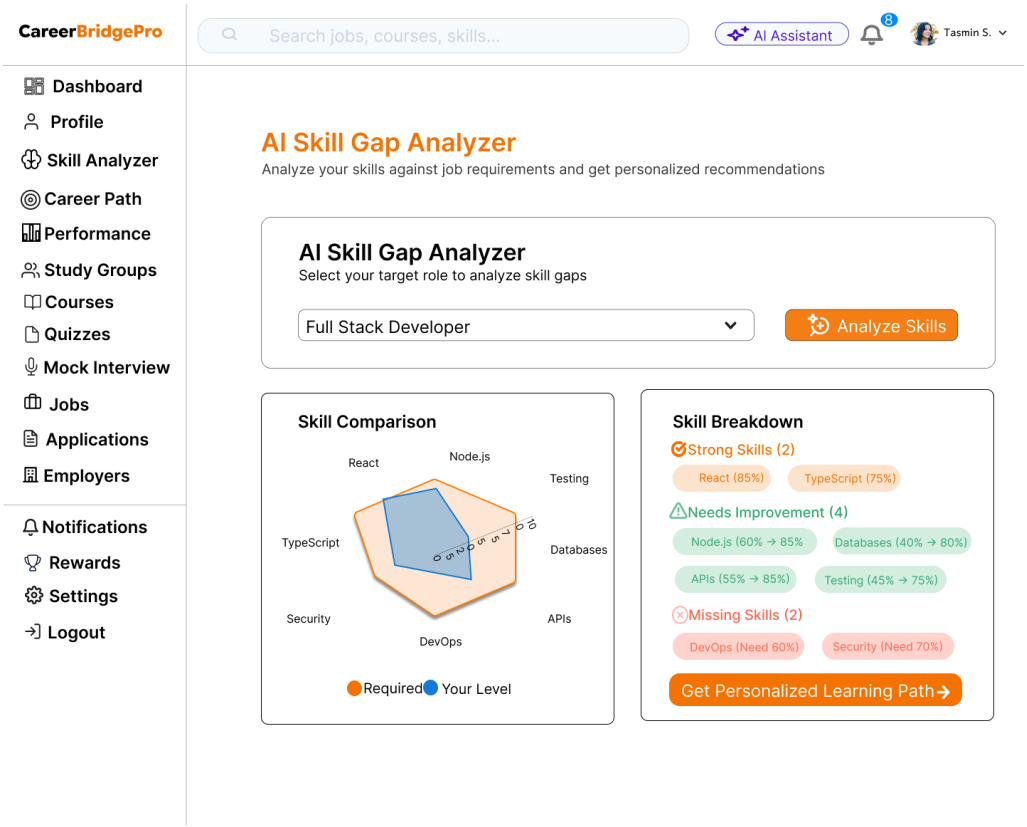


Figure 8.2: Desktop Skill Gap Analysis

Skill Gap Analysis

- Displays the user's current skills and compares them with career requirements.
- Identifies missing and weak skills.
- Provides AI-generated recommendations.
- Suggests courses and learning resources.
- Shows skill proficiency and progress.

8.9.2 Dashboard

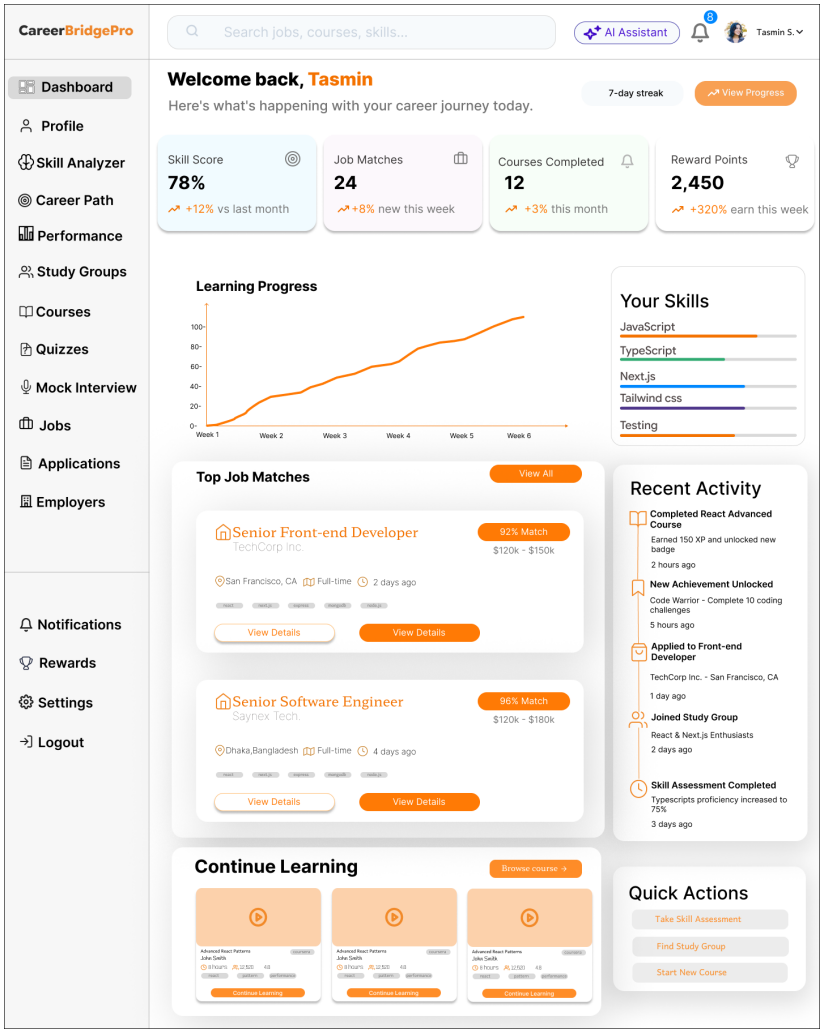


Figure 8.3: Desktop Dashboard

Dashboard (Home)

- Displays user progress and learning performance.
- Shows AI skill gap results and career goals.
- Job match percentages and recommendations.
- Notifications, badges, and achievements.

8.9.3 Mock Interview

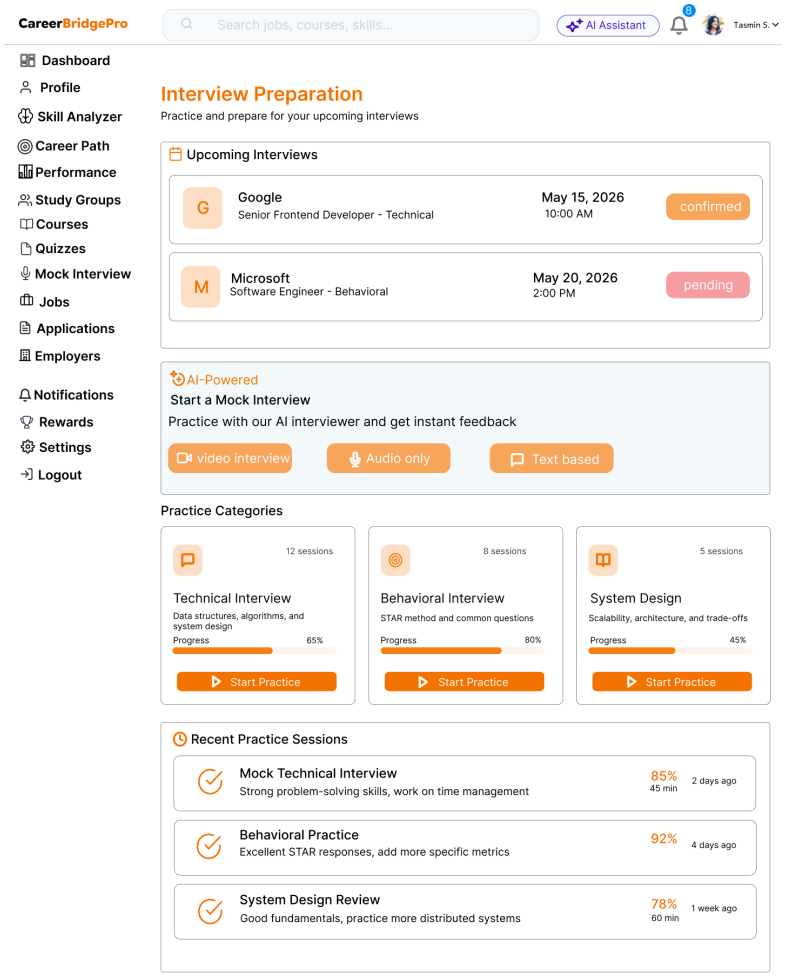


Figure 8.4: Desktop Mock Interview

Mock Interview

- Simulates real interview scenarios.
- Provides technical and behavioral questions.
- Generates performance scores and feedback.
- Identifies strengths and weaknesses.

8.9.4 Course Recommendations

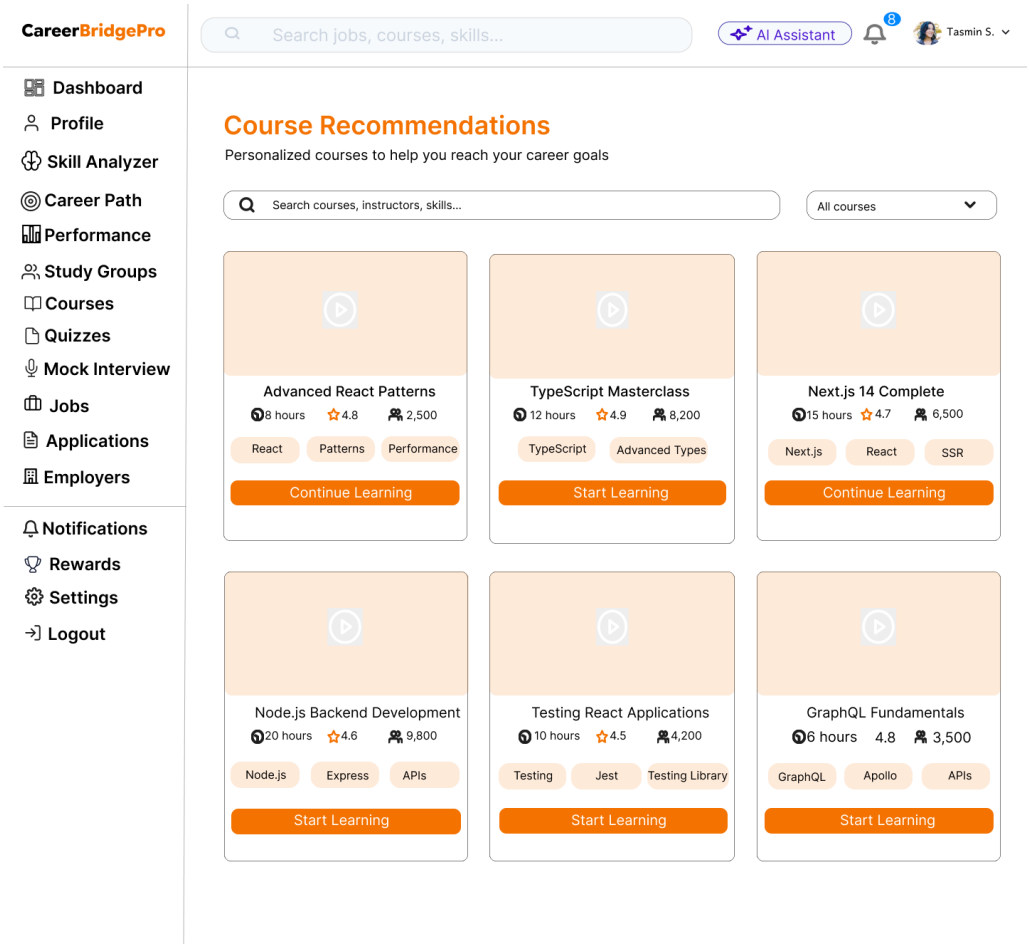


Figure 8.5: Desktop Course Recommendations

Course Recommendations

- Provides personalized course suggestions.
- Shows course details and difficulty levels.
- Tracks learning progress.
- Updates recommendations dynamically.

8.9.5 Performance Tracking

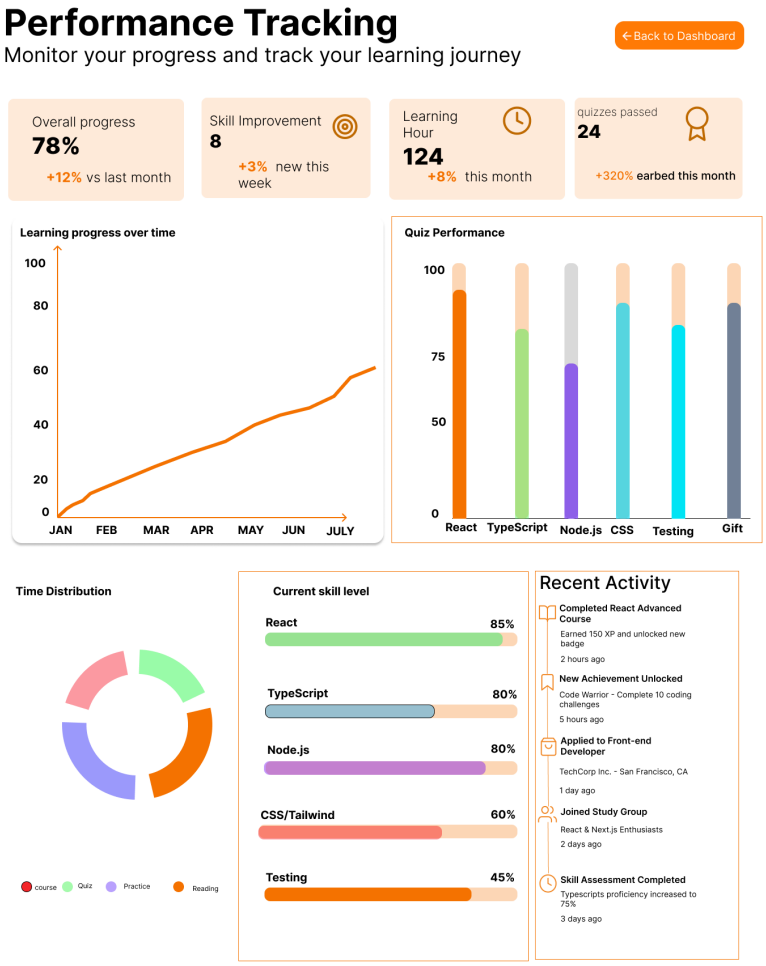


Figure 8.6: Desktop Performance Tracking

Performance Tracking

- Tracks learning progress and skills.
- Displays analytics and charts.
- Measures career readiness.
- Provides feedback and insights.

8.9.6 Rewards System

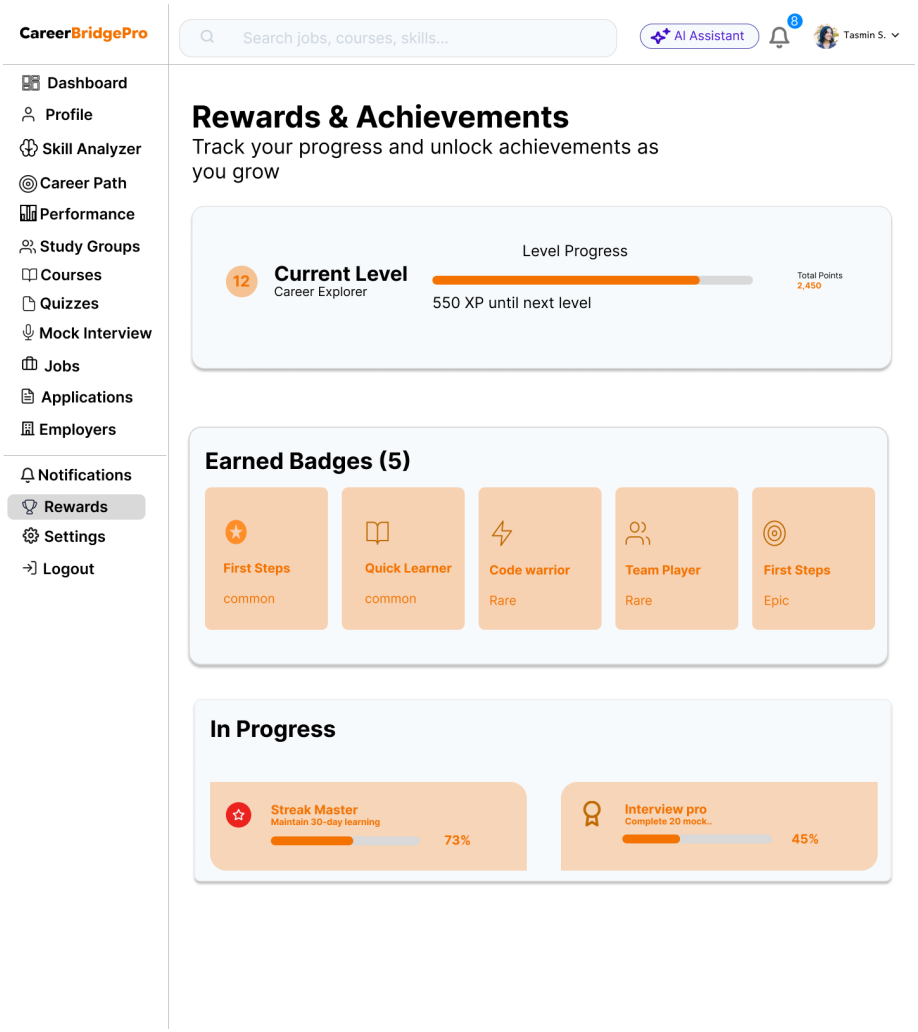


Figure 8.7: Desktop Rewards System

Rewards & Achievements

- Provides badges and XP points.
- Tracks achievements.
- Motivates continuous learning.
- Displays reward history.

8.9.7 Profile Management

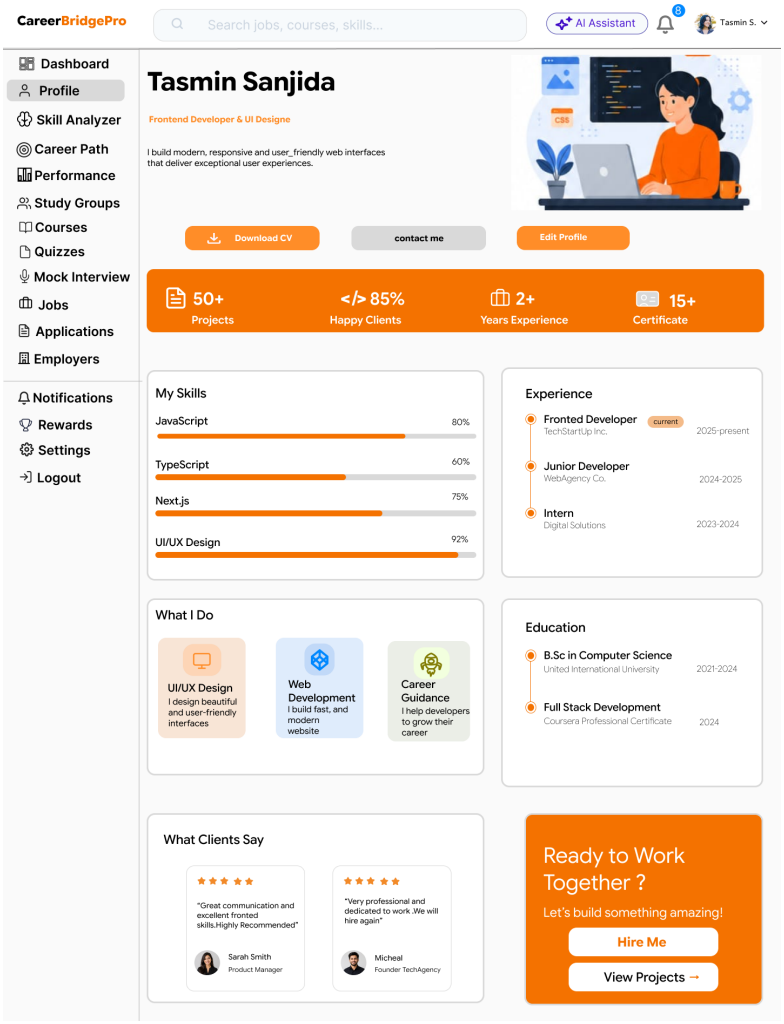


Figure 8.8: Desktop Profile Management

Profile Management

- Manages personal information and skills.
- Displays achievements and learning progress.
- Supports resume generation.
- Used for AI recommendations.

8.9.8 Career Path Planner

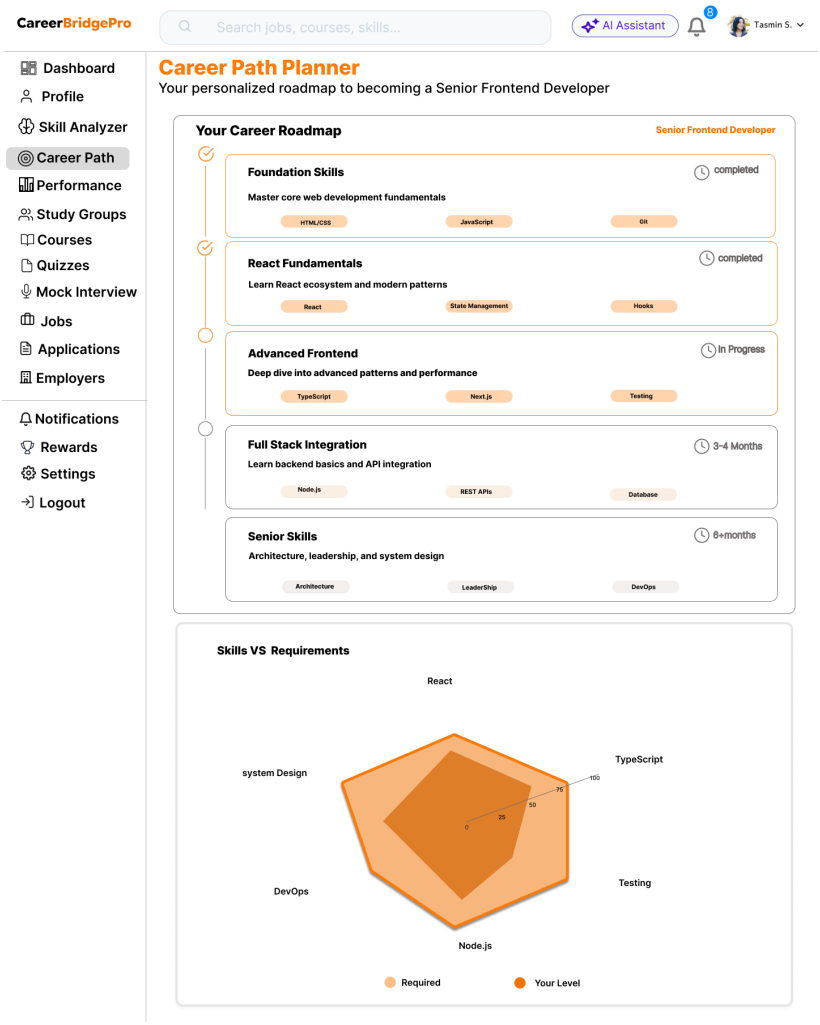


Figure 8.9: Desktop Career Path Planner

Career Path Planner

- Provides personalized career roadmap.
- Suggests skills and milestones.
- Tracks career progress.
- Motivates users.

8.9.9 Job Matching System

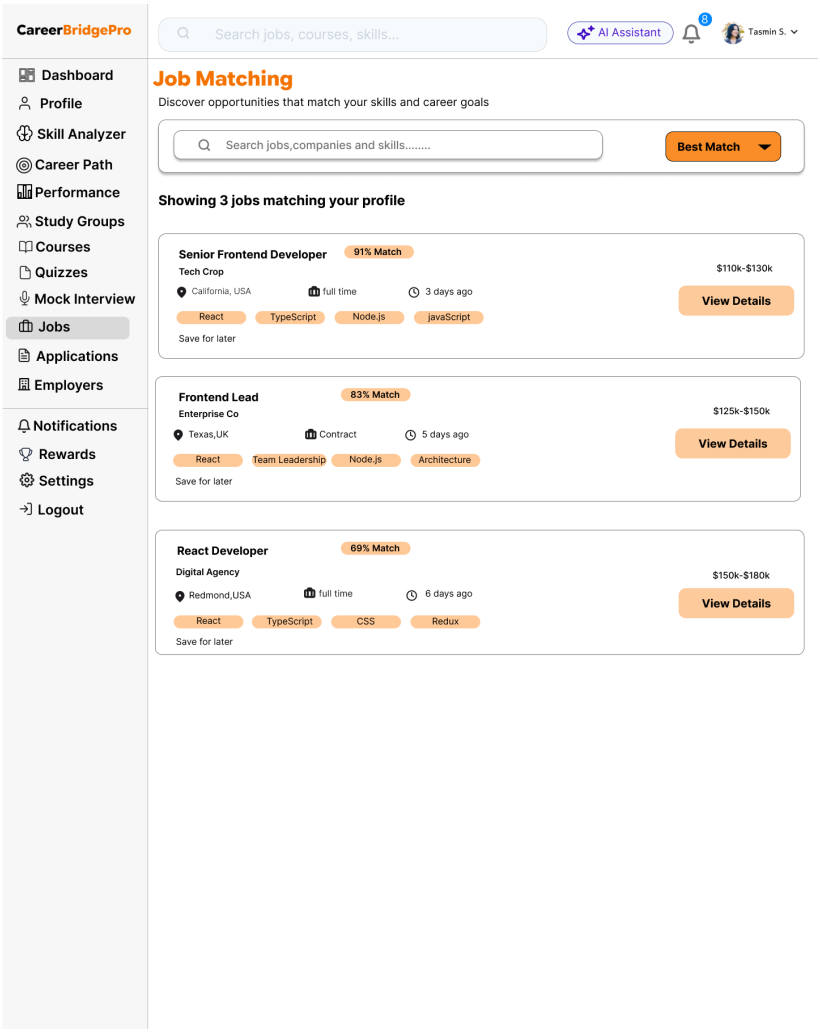


Figure 8.10: Desktop Job Matching System

Job Matching System

- Matches users with jobs.
- Shows match percentage.
- Highlights missing skills.
- Supports job applications.

8.9.10 Notifications & Activity

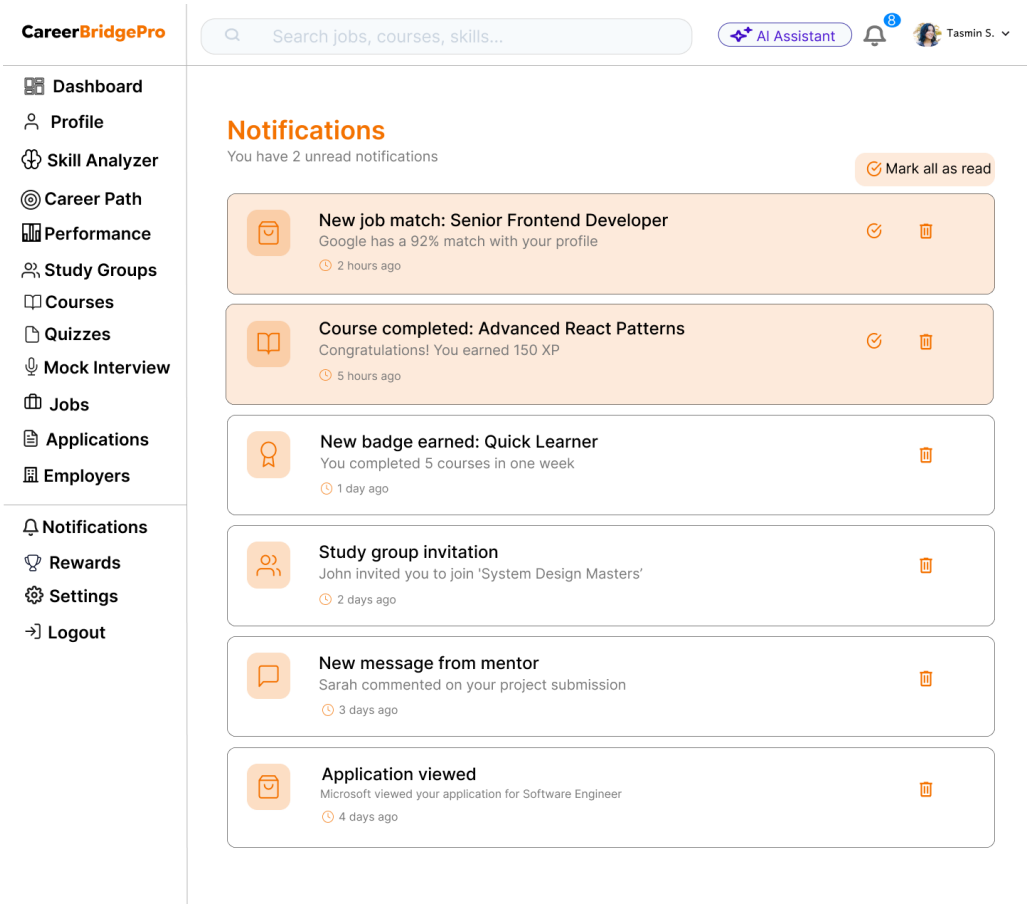


Figure 8.11: Desktop Notifications

Notifications & Activity

- Shows updates and alerts.
- Displays system activity.
- Keeps users informed.

8.10 Mobile Prototype Screens

8.10.1 Landing Page

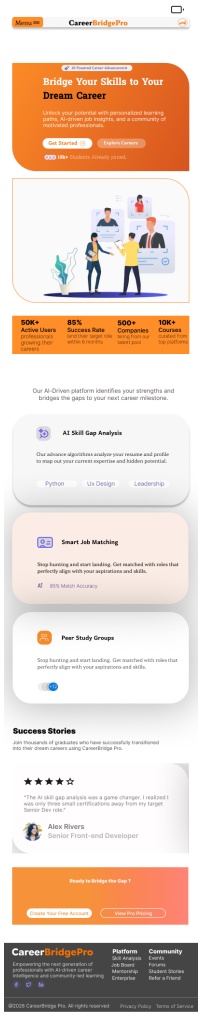


Figure 8.12: Mobile Landing Page

Mobile Landing Page

- Simplified introduction of CareerBridge Pro.
- Login and signup options.
- Key feature overview.

8.10.2 Skill Gap Analysis

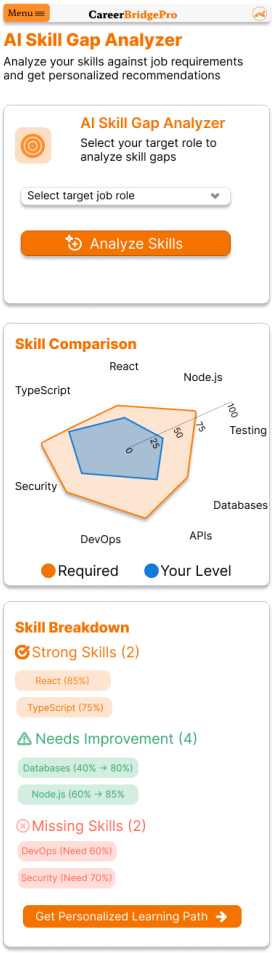


Figure 8.13: Mobile Skill Gap Analysis

Mobile Skill Gap Analysis

- Displays skill comparison.
- Shows missing skills.
- AI recommendations.

8.10.3 Dashboard

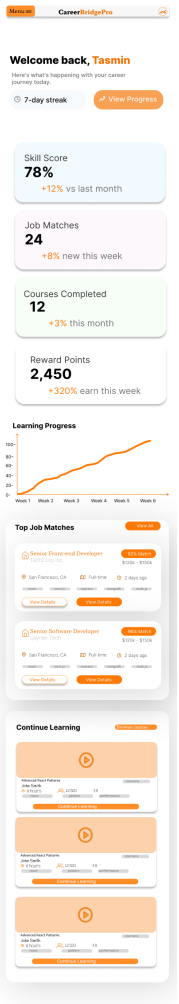


Figure 8.14: Mobile Dashboard

Mobile Dashboard

- Shows progress summary.
- Career insights.
- Quick navigation.

8.10.4 Performance

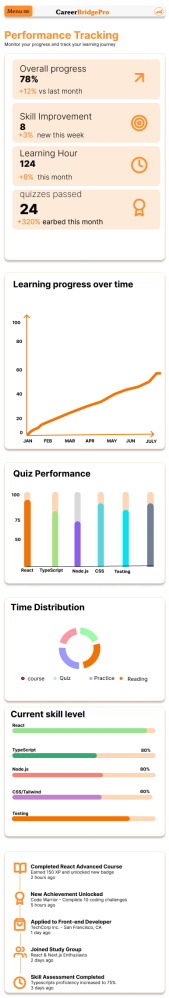


Figure 8.15: Mobile Performance Tracking

Mobile Performance Tracking

- Tracks learning progress.
- Displays performance charts.

8.10.5 Mock Interview

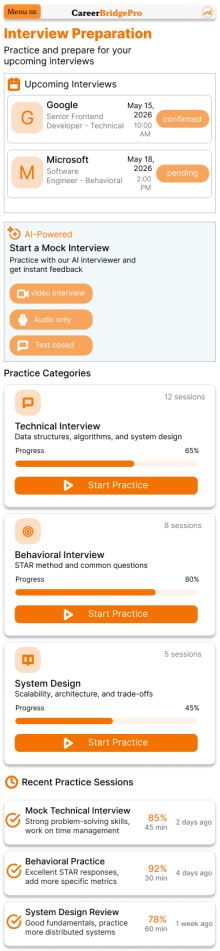


Figure 8.16: Mobile Mock Interview

Mobile Mock Interview

- Interview simulation.
- Feedback system.

8.10.6 Profile

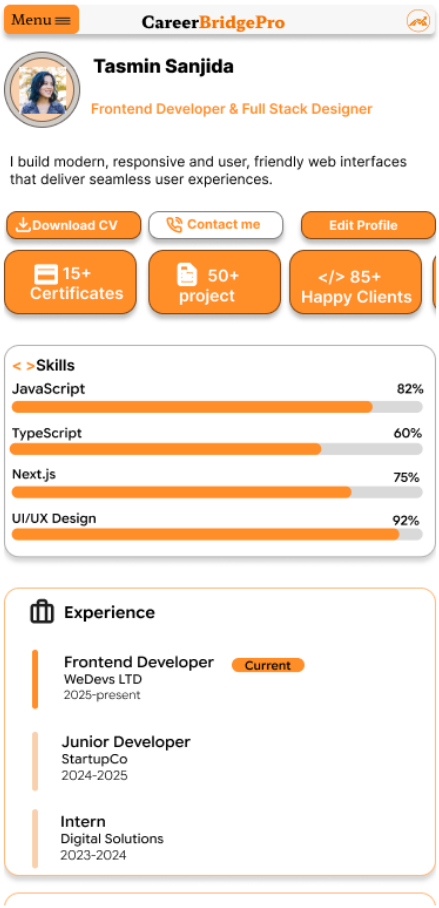


Figure 8.17: Mobile Profile

Mobile Profile

- Manages user profile.
- Shows achievements.

8.10.7 Employee View

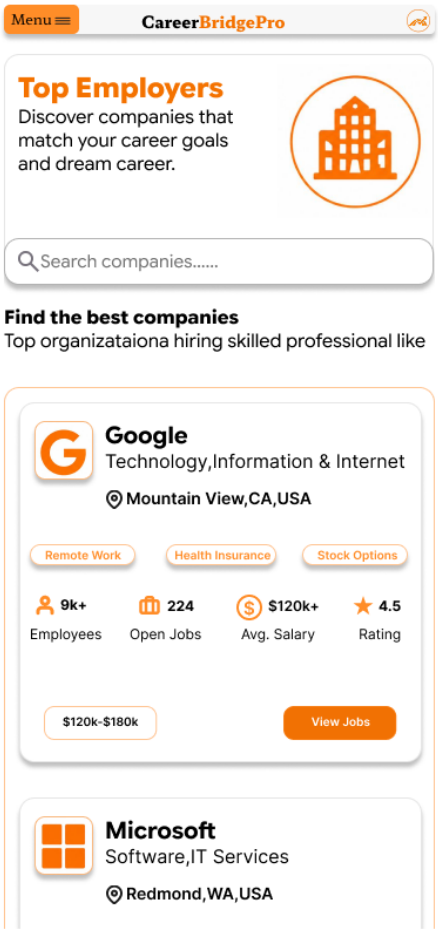


Figure 8.18: Mobile Employee View

Mobile Employee View

- Job-related interface.
- Simplified job browsing.

8.11 Navigation Flow

- Users start at Landing Page.
- After login, redirected to Dashboard.
- Sidebar navigation for all modules.
- Smooth transitions between sections.

8.12 Summary

The prototype design of CareerBridgePro ensures a modern, user-friendly, and scalable UI system that supports both desktop and mobile users effectively.

Chapter 9

Conclusion

In response to the evolving gap between education and industry needs, the CareerBridge Pro has been conceived as an intelligent solution that streamlines the entire learning to work experience. The platform will assist students and job seekers in recognizing their skill deficits, learning them and charting them over career path trajectories, while monitoring their learning progress and providing avenues of employment based on their newly acquired skills.

The platform is equally instrumental in enabling employers to efficiently identify and hire suitable candidates whose skills can add value to their organizations. With the integration of artificial intelligence powered skill analysis, a dedicated career roadmap, the ability to track and monitor the progression of individual users, and an efficient mechanism for matching individuals to the jobs they seek. CareerBridge Pro offers an all-encompassing learning to work environment.

This document for the Software Requirements Specification (SRS) for the platform has been meticulously crafted to detail the objectives of the system, define the problem statements and delineate the functional and non-functional requirements, along with defining the approach toward building this system. This will form the bedrock of all processes involved in the creation of the CareerBridge Pro, be it the designing, development, testing, and its eventual deployment and maintenance phase.

Essentially, CareerBridge Pro will enable both students and employers with better job prospects, learning capabilities, and hiring mechanisms respectively.

APPENDICES

Appendix A: Glossary

This glossary defines the key technical terms and acronyms used in the CareerBridge Pro System Analysis and Design document.

1. **AI (Artificial Intelligence):** A technology that enables systems to imitate human intelligence and offer smart recommendations, predictions, and decision-making support.
2. **API (Application Programming Interface):** A set of rules and protocols that lets different software applications talk to each other and share data.
3. **SDLC (System Development Life Cycle):** A structured process used for planning, designing, developing, testing, deploying, and maintaining a software system.
4. **DFD (Data Flow Diagram):** A graphic representation that shows how data moves through a system and how it is processed.
5. **UML (Unified Modeling Language):** A standard modeling language used to visualize and document software system design through diagrams like Use Case Diagrams and Activity Diagrams.
6. **Database:** An organized collection of data stored electronically and managed for easy access, modification, and retrieval.
7. **Frontend:** The user-facing part of the system that lets users interact with the application through visual interfaces.
8. **Backend:** The server-side component that handles business logic, databases, authentication, and system operations.
9. **Job Matching System:** A feature that compares user skills, qualifications, and preferences with job requirements to suggest suitable job opportunities.

10. **Skill Gap Analysis:** A process that finds missing or weak skills by comparing a user's current abilities with industry or job requirements.
11. **UI/UX (User Interface/User Experience):** UI refers to the visual design and interactive elements of the system, while UX focuses on overall usability and user satisfaction.
12. **Authentication:** The process of verifying a user's identity before granting access to the system.
13. **Profile System:** A feature that stores and manages user information like education, skills, achievements, and career goals.
14. **Recommendation System:** An intelligent system that suggests jobs, courses, learning resources, or career paths based on user data and preferences.
15. **Mock Interview System:** A feature that simulates real interview scenarios to help users practice and improve their interviewing skills.

Appendix B: Future Enhancements

Future versions of CareerBridge Pro may include the following features:

- AI-Powered CV/Resume Builder
- Video Interview Analysis
- Internship Recommendation System
- AI Career Counseling Chatbot
- Employer Performance Evaluation Dashboard
- Freelancing Opportunity Module
- Mentorship and Mentee Matching Program
- Mobile Application Version
- Global Job Matching and International Career Support
- Advanced Skill Certification Tracking
- Real-Time Industry Trend Analysis